Aerodynamic Characterization of Hollow Conical Shells in Free Fall

A Comparative Experimental and Numerical Study on the Influence of Apex Angle on Subsonic Drag Coefficients

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by

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Preface

The intersection of simplistic geometry and complex, non-linear wake topology presents a classic yet demanding aerodynamic challenge. The research described in detail on the following pages is the culmination of an experimental project carried out in 2024, in which I investigated the relationship between the internal angle of a cone and its terminal velocity.

This project was not only a significant milestone in my educational journey—resulting in me winning first place in the 20th Open Inter-School Physics Competition named after Bożena Koronkiewicz, but also a pivotal moment in my academic life. It was this specific research experience that solidified my conviction that Aerospace Engineering was the right path for me, directly inspiring my decision to apply to the Delft University of Technology. I carried out this research as a student at Secondary School No. 5 (V Liceum Ogólnokształcące) in Wrocław, Poland, under the supervision of my physics teacher, Ms Natalia Buczak.

This report considerably extends the initial experimental framework by introducing uncertainty propagation, computational fluid dynamics (CFD) validation, and refined analysis of the semi-empirical drag model, elevating the research to the rigorous standards of Delft University of Technology.

Kajetan R. Gułaj
Delft, January 2026

Abstract

This paper investigates the functional dependence of the aerodynamic drag coefficient (C_d) on the internal apex angle (α) of hollow conical shells falling in the subsonic regime ($Re \sim 10^4$). In this domain, flow separation is strictly forced by the sharp basal edge, effectively isolating geometric form drag from Reynolds number scaling effects. Five geometries $\alpha = 30°, 60°, 90°, 120°, 150°$ with constant mass and cross-sectional area were tested experimentally to determine their terminal velocities (v_t) and drag coefficients (C_d). Results demonstrate that C_d scales non-linearly with the apex angle, adhering to a semi-empirical formulation $C_d(\alpha) = a\sin^2(\frac{\alpha}{2}) + b$, ($R^2 \approx 0.99$). The strictly positive intercept ($b > 0$) empirically isolates a baseline base-pressure and viscous drag component, exposing a physical limitation of classical Newtonian impact theory.

Experimental findings were validated against Computational Fluid Dynamics (CFD) simulations using the $k-\omega$ SST turbulence model. The numerical predictions showed excellent agreement for blunt geometries (deviation $\lesssim 2\%$), but significantly underestimated drag for the slender 30° cone (−27%). This divergence is attributed to fluid-structure interactions (flutter) in the non-rigid experimental sample, which are not captured by steady-state RANS formulations. The study concludes that for hollow cones, the Newtonian impact theory is insufficient, and a modified model accounting for base pressure and wake topology is required to accurately predict descent velocity.

Contents

List of Symbols

A Reference surface (area perpendicular to the direction of movement). 1, 2, 4, 5, 9, 12, 16, 17, 26, 34

a Empirical slope parameter in the semi-empirical model $C_d = a \sin^2(\alpha/2) + b$. 2, 28, 34, 48

α Internal apex angle of a cone. 2, 3, 5, 8, 9, 11, 12, 15–18, 20, 21, 23, 24, 27, 34, 36

b Empirical intercept parameter in the semi-empirical model $C_d = a \sin^2(\alpha/2) + b$. 2, 28, 34, 48

C_d Drag coefficient. 1–9, 11–13, 16–18, 20–25, 27, 28, 33, 34, 36

F_D Aerodynamic drag force. 4

F_g Gravitational force (weight). 4

g Gravitational acceleration. 4

m Mass of the object. 2, 4, 5, 9, 16, 17, 34

Ma Mach number. 5

Re Reynolds number. 3, 5, 25

ρ Density of a medium (e.g. air). 2, 4, 5, 11, 17, 34

v_t Terminal velocity. 2–5, 9–12, 16–18, 25, 34–36

List of Abbreviations

***k*–ϵ** k–ϵ turbulence model. 20, 21

CFD Computational Fluid Dynamics. 2, 3, 7, 18, 21–25, 27, 28, 30–33, 36, 48, 49

OLS Ordinary Least Squares. 30–32, 48, 49

PIV Particle Image Velocimetry. 37

RANS Reynolds-Averaged Navier–Stokes. 18–21, 23–25, 27, 37

SE Standard Error. 15

SIMPLE Semi-Implicit Method for Pressure Linked Equations. 18

SST k–ω Shear Stress Transport. 19, 21, 23, 25–27

URANS Unsteady Reynolds-Averaged Navier–Stokes. 37

WLS Weighted Least Squares. 28–30, 32, 36, 48, 49

1
Introduction

This chapter establishes the foundational context and outlines the core motivation for investigating the aerodynamic characteristics of hollow conical shells in free fall. First, section 1.1 introduces the fundamental physics of aerodynamic drag and terminal velocity, leading to the formulation of the primary research question and the specific objectives of the study (detailed in section 1.1.1). Finally, section 1.2 defines the physical, kinematic, and numerical boundaries of the research, clearly delineating the scope and limitations of the applied methodologies.

1.1. Research Problem and Objectives

Aerodynamic drag is a dissipative force opposing the relative motion of a body through a fluid. For macroscopic objects moving through air at moderate subsonic speeds, this force is effectively approximated by a quadratic velocity law. In this regime, the magnitude of the resistance force depends on the density of the medium, the velocity, the reference surface area, and a dimensionless drag coefficient that captures the complex dependencies on the geometry of the object and the flow conditions [2].

This relationship is written as

$$F_d = \frac{1}{2}\rho C_d A v^2, \tag{1.1}$$

where C_d is an empirical parameter, usually determined experimentally (e.g., in a wind tunnel), and its value may vary significantly depending on the shape and flow regime.

Shape dependence is critical in fluid dynamics; bodies with identical frontal areas (A) exhibit drag coefficients (C_d) varying by orders of magnitude depending on flow separation and wake topology. For example, a comparison of typical C_d values for several solids shows that blunt shapes (e.g., a flat plate) have notable higher drag than streamlined shapes (e.g., a profile), and modifying the geometry at the rear of an object can reduce drag (e.g., comparison of a sphere and a bullet shape). This relationship is illustrated in fig. 1.1.

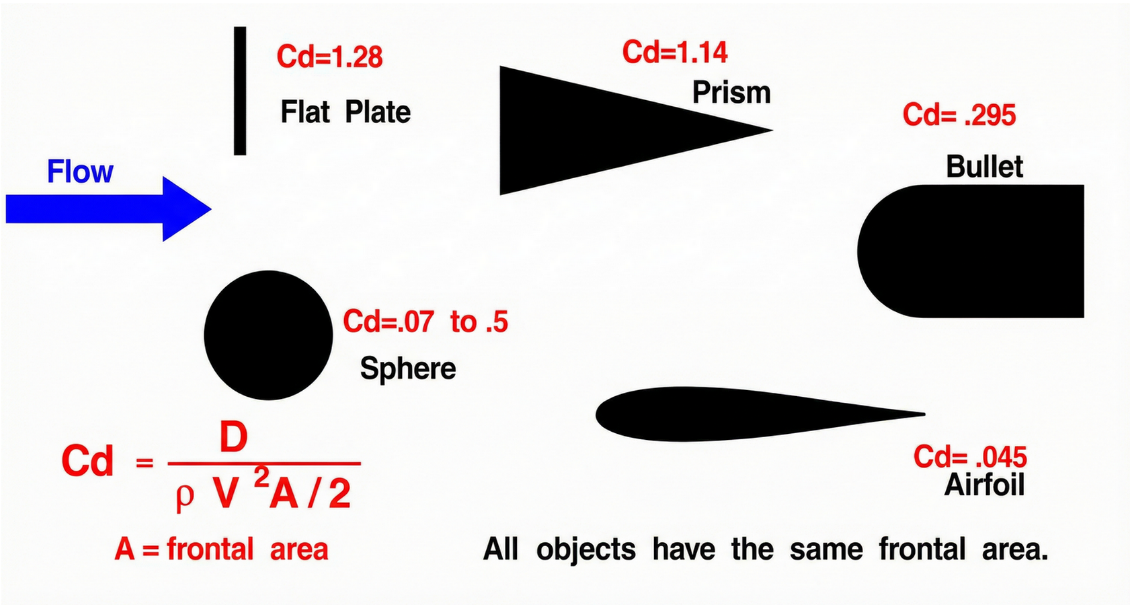


Figure 1.1: *Influence of shape on typical C_d values for the same reference field [3] (Modified).*

In the context of vertical motion, an important concept is the terminal velocity, which is reached when the force of gravity balances aerodynamic drag (ignoring the usually small force of buoyancy in the air). In a steady state (no acceleration), the condition of force equilibrium is met, and after substituting eq. (1.1), we obtain

$$v_t = \sqrt{\frac{2mg}{\rho A C_d}}. \tag{1.2}$$

It follows directly from this that, with constant m, ρ and A, the terminal velocity is inversely proportional to the square root of C_d, i.e. $v_t \propto \frac{1}{\sqrt{C_d}}$.

This work focuses on a quantitative description of how cone geometry influences aerodynamic drag and the resulting descent velocity. The primary research question is:

How does the internal apex angle α govern the terminal velocity v_t and the drag coefficient C_d?

This inquiry addresses the classic "falling paper cones" problem, a well-known benchmark in fluid dynamics used to investigate the relationship between bluff-body geometry and steady-state descent [4]. To answer this, the study aims to determine the function $C_d(\alpha)$ through both physical experimentation and numerical simulation Computational Fluid Dynamics (CFD). A key aspect of this analysis is to assess the adherence of observed descent velocities to the theoretical inverse-square root law ($v_t \propto 1/\sqrt{C_d}$) derived from the standard drag equation eq. (1.2).

1.1.1. Objectives of the Study

The primary goal is to derive a mathematical description of the cone's aerodynamic behavior in the quadratic drag regime [2], [5]. To achieve this, the project is structured around six specific objectives:

O1. Characterize the Drag-Angle Relationship $C_d(\alpha)$. Determine the functional relationship between the cone's apex angle α and its drag coefficient C_d. This coefficient serves as the empirical measure of how shape and flow regime dictate resistance.

O2. Verify Velocity Dependence. Confirm that for a system with constant mass and reference area, the terminal velocity follows the theoretical equilibrium condition:

$$v_t \propto \frac{1}{\sqrt{C_d}}$$

This step ensures that observed velocity changes are driven purely by aerodynamic factors rather than mass or area discrepancies.

O3. Conduct Controlled Experimental Validation. Measure the terminal velocity v_t for a discrete set of five cones ($\alpha \in \{30°, 60°, 90°, 120°, 150°\}$). Crucially, this experiment maintains a strictly constant mass and projected base area, isolating geometry as the sole independent variable [4].

O4. Corroborate via Computational Fluid Dynamics CFD. Perform numerical simulations to validate the experimental data and visualize the underlying flow physics. Special attention is paid to the wake structure and separation region to explain deviations from simplified theoretical models.

O5. Formulate a Semi-Empirical Model. Evaluate the limitations of the standard Newtonian approximation ($C_d \propto \sin^2(\alpha/2)$) and formulate a corrected semi-empirical model accounting for base drag and effective bluntness;

$$C_d = a \sin^2\left(\frac{\alpha}{2}\right) + b$$

Here, the term proportional to a captures the geometry-dependent effective bluntness, while the constant b accounts for the baseline drag contribution from viscosity and wake formation.

O6. Quantify uncertainty and classify the sources of variability. Conducting a full uncertainty analysis, including the propagation of measurement uncertainties through non-linear relationships using the aerodynamic drag equation and the relationship to the terminal velocity.

1.2. Scope and Limitations

The scope of this report investigates the aerodynamics of low-rigidity, thin-walled hollow conical shells falling in a vertex-down orientation under subsonic flow conditions, which are treated as incompressible in the speed range under consideration. Consequently, the analysis is based on the classical description of square drag and on the determination of the drag coefficient C_d and the terminal velocity v_t for this specific geometric and kinematic configuration [2].

A distinction exists between solid and hollow cones, unlike solid cones, hollow shells enforce flow separation at the sharp basal edge, generating a significant pressure drag component dominated by the wake structure, which has a direct impact on the observed C_d value. [6, pp. 75–89], [7], [8]

This paper only considers sections of flow where the descent is stable or quasi-stable (i.e. without sudden changes in orientation and without strong oscillations). This approach is necessary because for non-spherical bodies, the orientation in the flow may depend both on the geometry (e.g., the apex angle α) and on the inertial-viscous regime, and in certain ranges, orientation switching, tilted states, or oscillations related to wake dynamics may occur. The literature shows that cones can exhibit stable or oscillating orientation behaviour, depending on the apex angle and Reynolds number Re, which justifies the selection of only those data fragments that correspond to steady (or near-steady) descent.[4], [9]

The numerical scope of this study is limited to steady-state Reynolds-Averaged Navier-Stokes (RANS) simulations, utilized to validate experimental findings and visualize steady flow structures. The Computational Fluid Dynamics CFD analysis employs the k-ω SST model as the primary closure for turbulence, with the standard k-ϵ model applied for sensitivity analysis specifically regarding the anomalous behaviour of the slender 30° geometry. In alignment with the physical experiment, the numerical domain assumes incompressible flow, neglecting compressibility effects which are insignificant at the observed descent velocities.

An additional limitation is the range of dynamic similarity: in this study, the range of interest for the Reynolds number corresponds to flows of the order of

$$10^4 \lesssim Re \lesssim 3 \times 10^4,$$

which corresponds to typical conditions where the drag coefficient C_d may depend on Re, but at the same time modelling based on averaged values and parameterisation via C_d is used. In particular, cases of low Reynolds numbers (viscous regime with linear resistance) and flows in which compressibility effects become notable are not analysed. [8] [6, pp. 75–89]

Finally, the results and conclusions refer exclusively to hollow cones falling with their apex pointing downwards, without deliberate rotation, and for the experimental conditions described in the following chapters. Any extrapolations to other configurations (e.g., cone with its apex facing upwards, cone in an inclined position, solid cones, or other rotating solids) are beyond the scope of this work and require separate experimental validation.

2

Theoretical Framework

In order to accurately interpret experimental data, it is necessary to establish the physical laws governing fluid resistance. This chapter begins with the basic equation of air resistance and then moves on to the detailed aerodynamics of conical shapes, comparing the historical Newtonian model with modern fluid dynamics theory.

2.1. Physics of Terminal Velocity

The motion of a body falling through the air is determined by the interplay of two main forces acting along the vertical axis: the force of gravity F_g (directed downwards) and the force of aerodynamic drag F_D (directed opposite to the instantaneous velocity vector). In a typical description for macroscopic objects moving through the air at moderate speeds, it is assumed that aerodynamic drag is approximately proportional to the square of the velocity v, and its value depends on the density of the medium ρ, the reference area A (for a cone falling with its apex downwards; $A = \pi r^2$), and the drag coefficient C_d, which effectively parameterises the influence of shape and flow conditions [2], [5].
Thus, along the direction of motion, the drag equation can be written as

$$F_d = \frac{1}{2} \rho C_d A v^2, \tag{2.1}$$

The gravitational force for a body with mass m in a uniform gravitational field with acceleration g is given by

$$F_g = mg. \tag{2.2}$$

During the initial descent at low speeds, drag F_d is negligible, and the body accelerates downwards. However, as the speed increases, F_d increases (according to eq. (2.1)), which reduces the acceleration. The terminal velocity v_t is reached when the resultant force along the direction of motion disappears, i.e., when the drag force balances the weight. Figure 2.1 illustrates the asymptotic approach to equilibrium velocity (v_t) as aerodynamic drag counteracts gravitational acceleration.

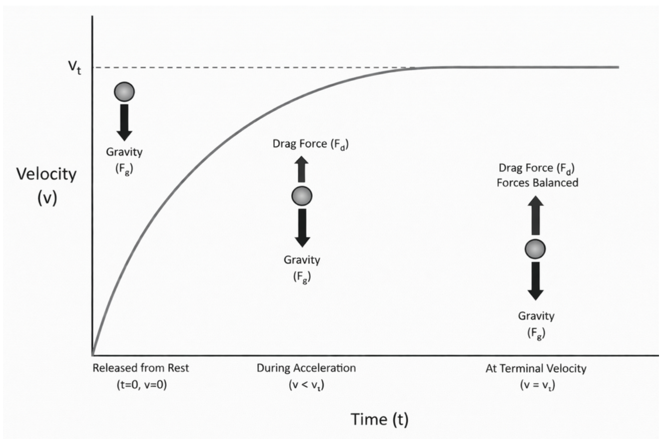


Figure 2.1: *The graph illustrates the velocity-time curve and the point where equilibrium (terminal velocity) is reached (Modified [5]).*

Assuming a positive downward convention and neglecting buoyancy effects as the density ratio $\rho_{\text{air}}/\rho_{\text{cone}} \ll 1$, rendering the buoyant force statistically insignificant compared to gravitational and drag forces, the steady-state condition can be written as

$$F_g - F_d = 0 \quad \Rightarrow \quad mg = \frac{1}{2} \rho C_d A v_t^2. \tag{2.3}$$

Solving eq. (2.3) with respect to v_t yields the classical expression for the terminal velocity in the quadratic drag regime:

$$v_t = \sqrt{\frac{2mg}{\rho\, C_d\, A}}. \tag{2.4}$$

From eq. (2.4) it follows directly that, for constant parameters m, ρ, and A, the terminal velocity v_t is inversely proportional to the square root of C_d, i.e. $v_t \propto \frac{1}{\sqrt{C_d}}$. This relationship forms the basis for further experimental analysis: measuring v_t under controlled conditions allows C_d to be determined (after transforming eq. (2.4)) and the obtained values to be compared with models of the relationship between C_d and cone geometry.

2.2. Newton's Impact Theory

To derive a theoretical baseline for the relationship between the cone's geometry and its drag coefficient, this section explores the classical Newtonian impact theory. While originally formulated for idealized corpuscular dynamics, its simplified approach provides a foundational mathematical framework that explicitly links the apex angle α to the expected drag force. The subsequent subsections detail the assumptions of this model and derive the geometric drag dependency.

2.2.1. Drag coefficient in respect to geometry of the cone

The drag coefficient C_d is a variable function of geometry, Reynolds number (Re), and Mach number (Ma), rather than a fundamental constant [8]. Under the conditions of this experiment, the velocities are significantly lower than the speed of sound in air ($v \ll 340\,\mathrm{m/s}$), therefore the effect of compressibility is negligible and Ma ≈ 0.01. Furthermore, for objects with sharp edges in a regime of sufficiently high Reynolds numbers (of the order of $Re > 10^4$), the drag coefficient is approximately independent of Re, because the separation point is then largely determined geometrically; in particular, for sharp-cornered bodies (so-called *sharp-cornered bluff bodies*), the dependence on Re may become negligible [8], [10]. Consequently, in the analysed system (an empty cone falling with its apex downwards and a sharp edge at its base), C_d can be treated as a function of the geometry itself in the first order of approximation.

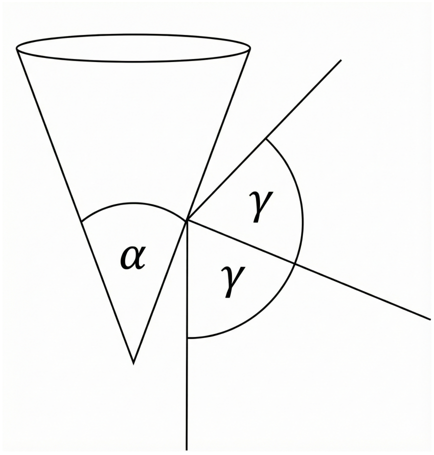


Figure 2.2: *The geometry of a cone in a mathematical model: definition of the apex angle α and angles used in the derivation. (generated with help of AI)*

2.2.2. Assumptions of Newtonian model

The derivation relies on a simplified corpuscular flow model, treating the fluid as a stream of non-interacting particles moving in a straight line at a constant speed v relative to a stationary cone. [4], [11]
It is assumed that:
(i) collisions of particles with the cone surface are perfectly elastic,
(ii) particles do not interact with each other (no particle-particle collisions),
(iii) the flow is uniform and linear (no viscosity, no boundary layer, no turbulence and no wake),
(iv) the drag force results solely from the momentum balance in the direction of flow.
In this approach, resistance is interpreted as the result of a change in the momentum component of particles reflected from the surface.

2.2.3. Change in momentum and drag force

One can consider an element of the surface area dA of a cone, which is struck by particles of mass m and velocity v during time dt. If γ denotes the local angle between the direction of the stream and the normal to the surface, then in an elastic collision the sign of the normal component of the velocity changes, leading to a change in the momentum of a single particle along the axis of flow proportional to $\cos^2(\gamma)$, while the transverse components average out to zero due to the rotational symmetry of the cone [4], [11].

The number of particles hitting dA in time dt is proportional to the particle flux and the component of velocity perpendicular to the surface, i.e. to $v\cos(\gamma)$. After summing the contributions (momentum change × number of collisions) and using the medium density relation $\rho = Nm$ (where N is the number of particles per unit volume), we obtain the elementary resistance force acting on dA in the form of

$$dF = 2\rho v^2 \cos^3(\gamma)\, dA. \tag{2.5}$$

Integrating eq. (2.5) over the entire lateral surface of the cone, we initially obtain an expression for the drag force in general form:

$$F_d = 2\rho v^2 \iint_S \cos^3(\gamma)\, dA. \tag{2.6}$$

To solve this integral, the geometric factor I_{geo} must be determined. Assuming that the slant height of the cone is described by the linear function $y(x) = ax$, where the derivative $y'(x) = a$ is constant, the integral over the radius r can be written and calculated as follows [11]:

$$\frac{I_{\text{geo}}}{\pi} = \int_0^r \frac{2x}{1+y'(x)^2}\, dx = \int_0^r \frac{2x}{1+a^2}\, dx = \left[\frac{x^2}{1+a^2}\right]_0^r = \frac{r^2}{1+a^2}. \tag{2.7}$$

Knowing that the cross-sectional area is $A = \pi r^2$, and using trigonometric identities, this fraction can be rewritten as:

$$\frac{1}{1+a^2} = \frac{1}{1+\tan^2(\gamma)} = \cos^2(\gamma). \tag{2.8}$$

Subsequently, taking into account the relationship between the angle of inclination of the wall γ and the internal apex angle of the cone α, which is $\gamma = \frac{\pi}{2} - \frac{\alpha}{2}$, we obtain the final form of the geometric factor:

$$\cos^2(\gamma) = \cos^2\left(\frac{\pi}{2} - \frac{\alpha}{2}\right) = \sin^2\left(\frac{\alpha}{2}\right) \implies I_{\text{geo}} = A\sin^2\left(\frac{\alpha}{2}\right). \tag{2.9}$$

Substituting the designated geometric factor I_{geo} back into the force equation, we obtain the final formula:

$$F_d = 2\rho v^2 A \sin^2\left(\frac{\alpha}{2}\right). \tag{2.10}$$

2.2.4. Drag coefficient derived from the Newtonian model

Comparing eq. (2.10) with the standard form of the drag equation eq. (2.1), the drag coefficient C_d for the cone is obtained directly:

$$C_d = 4\sin^2\left(\frac{\alpha}{2}\right). \tag{2.11}$$

The result eq. (2.11) is attractive due to its simple, straightforward dependence on geometry (only on α), but it should be remembered that it is a consequence of adopting highly idealised assumptions: no viscosity, no interactions between particles, and no turbulent aerodynamic wake [11]. It is worth noting here that the standard Newtonian theory of flow, often used in hypersonic aerodynamics, assumes inelastic collisions (where the normal component of momentum is lost rather than reversed), which leads to a coefficient of 2 instead of 4. However, regardless of whether the theoretical factor is 2 or 4, the fundamental functional dependence on $\sin^2(\frac{\alpha}{2})$ remains unchanged, and both models share the same limitation. In the following sections, it will be shown that these limitations are crucial for cones falling in air and lead to the need to modify the model [6, pp. 877–881].

2.3. Limitations of the Newtonian Model and Semi-Empirical Correction

Although the Newtonian formulation derived in section 2.2 offers an elegant algebraic link between geometry and drag, it is fundamentally constrained by its frictionless, discrete-particle assumptions. This section critically examines these theoretical shortcomings in the context of subsonic, continuum fluid dynamics. Recognizing the discrepancies between analytical predictions and physical reality, a semi-empirical correction is subsequently proposed to bridge the gap between idealized impact theory and actual aerodynamic phenomena.

2.3.1. Predictions of the simplified Newtonian model

The derivation in eq. (2.11) leads to the relationship

$$C_d = 4\sin^2\left(\frac{\alpha}{2}\right),$$

and therefore predicts a range of values $C_d \in [0, 4]$, because $\sin^2(\frac{\alpha}{2}) \in [0, 1]$. This range alone indicates a contradiction with aerodynamic data for real solids in subsonic flow: typical values of C_d for objects with the same reference area can be, for example, approximately 1.28 for a flat plate positioned perpendicular to the flow and approximately 0.045 for a streamlined symmetrical profile.[3] Newton's model also allows for the limit $C_d \to 0$ for $\alpha \to 0$, which implies the disappearance of drag in the case of a 'very slender' cone, whereas in actual flow, drag cannot disappear due to non-zero viscosity and the associated surface friction.[6, pp. 75–89]

2.3.2. Physical Limitations of the Corpuscular Approximation

Newton's corpuscular theory fails to accurately predict subsonic drag because it fundamentally neglects the continuum properties of the fluid: viscosity and downstream interaction. First, the model incorrectly assumes a pressure-less wake region, neglecting the base pressure recovery and recirculation zones inherent to subsonic bluff-body flows [8]. Second, it ignores skin friction, which is a non-negotiable force, especially for slender bodies, and fails to account for the upstream stagnation field where streamlines deflect before impact [6, pp. 75–89]. Consequently, analytical algebraic equations are insufficient; Computational Fluid Dynamics (CFD) is required to resolve these complex viscous and pressure distributions.

2.3.3. Modified hypothesis: semi-empirical model

To address these physical omissions while retaining the geometric intuition of the collision model, a semi-empirical approach is proposed [6, pp. 75–89], [4]. Hypothesizing that total drag consists of a geometry-dependent pressure component (bluntness) and a baseline viscous component, the Newtonian parameter $\sin^2\left(\frac{\alpha}{2}\right)$ is retained as the scaling factor. The modified relationship is defined as:

$$C_d = a\sin^2\left(\frac{\alpha}{2}\right) + b, \tag{2.12}$$

where the slope a characterizes the increase in pressure drag and separation width with apex angle, while the intercept b represents the baseline drag (friction and base pressure) that persists even as $\alpha \to 0$, preventing the non-physical zero-drag prediction [4].

3

Methodology and Experimental Setup

This chapter details the experimental methodology employed to investigate the aerodynamic characteristics of the conical shells. It outlines the design and fabrication of the test specimens in section 3.1, describes the experimental apparatus and environmental conditions in section 3.2, and defines the measurement protocols used for data acquisition in section 3.3. Finally, an error analysis and uncertainty propagation framework is presented in section 3.4 to ensure the reliability of the empirical findings.

3.1. Design and Fabrication of Test Specimens

To systematically isolate the effect of the internal apex angle α on the drag coefficient, a set of customized physical models was required. This section provides an overview of the fabricated test specimens and details the control of geometric and mass parameters necessary to ensure a valid comparative analysis.

3.1.1. Overview of test specimens

As part of the experiment, five thin-walled, hollow conical shells were made of paper, with internal apex angles α belonging to the set $\{30°, 60°, 90°, 120°, 150°\}$. Each cone was designed to test aerodynamic drag in a free-fall configuration with the apex pointing downwards, so that the determined value of C_d would refer to comparable flow conditions. The set of samples is shown in fig. 3.2.

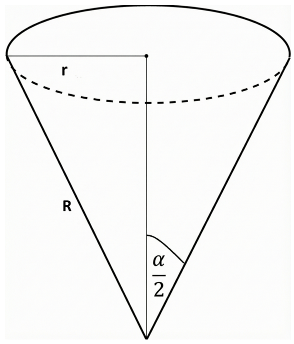


Figure 3.1: *Visual representation of parameters r, R and* α

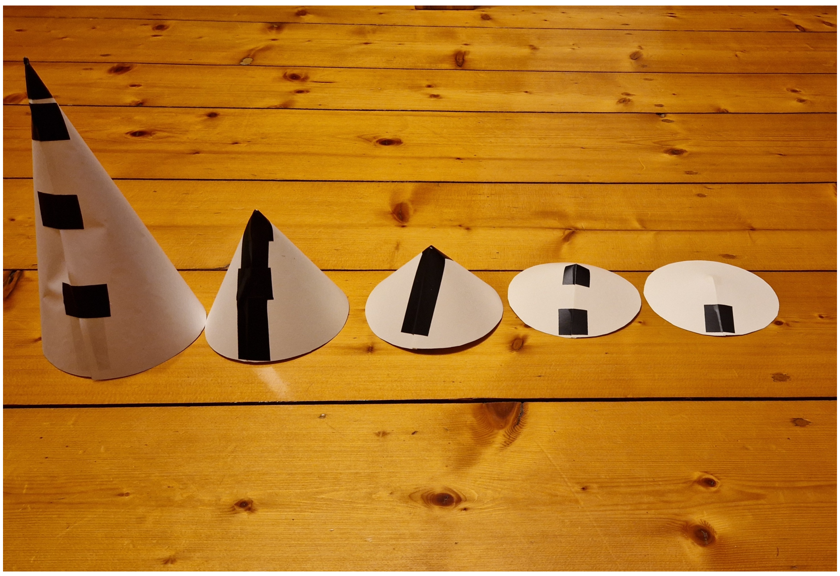


Figure 3.2: *Set of five hollow cones used in the study (from left:* 30°, 60°, 90°, 120°, 150°*).*

3.1.2. Geometric parameters and isolation of the influence of the internal apex angle

The key premise of the project was the geometric isolation of the independent variable: the only parameter differentiating the samples was to be the angle α. To this end, all cones were made with the same base radius $r = 6.0 \pm 0.1$ cm, which unambiguously imposed a constant reference surface area A for the tip-down configuration. The constancy of r was important because in the aerodynamic drag equation A appears as a factor linearly scaling the drag force, and thus directly affects the determined C_d. [2]

Thickness and material. In order to limit the variability resulting from design parameters, a constant material thickness was adopted for most samples: four cones (with angles of 60°, 90°, 120° and 150°) were made of poster board with a paper density of 200 $\mathrm{g/m^2}$ and a thickness of 0.3 mm. While the primary objective was to isolate α as the sole independent variable, the extreme aspect ratio of the 30° cone necessitated a structural compromise. To prevent macroscopic buckling, this specific geometry was fabricated from a lower-density material (125 $\mathrm{g/m^2}$) and a thickness of 0.2 mm.

Mass and reference area control (variable isolation). To ensure that α is the only independent variable, the key quantities determining the balance of forces were deliberately *fixed and standardised* for all samples:

Each cone was weighted to an equal mass of $m = 5.00 \pm 0.01$ g. The mass was balanced by placing a small amount of plasticine inside the apex, which also increased the stability of the orientation during the fall. Additionally, the mass introduced locally by the connecting tape was compensated for with plasticine on the opposite side to limit the eccentricity of the centre of mass. As a result, the force driving the fall (weight) was identical for all geometries:

$$F_g = mg. \tag{3.1}$$

Due to the constant base radius, the frontal projection area (base area) $A = 0.01131 \pm 0.00038\,\mathrm{m^2}$. The constancy of A eliminates the influence of the 'frontal' scale of the object on drag and allows differences in aerodynamic behaviour to be attributed solely to changes in geometry (angle α).

Since m and A are constant for all samples, and the flow conditions in the experiment correspond to the quadratic drag regime, any differences in the measured terminal velocity v_t must result directly from changes in C_d caused solely by the geometry of the cone. This relationship follows from the equation for terminal velocity in the quadratic drag regime, which indicates that for constant m and A, the terminal velocity satisfies the relation $v_t \propto \frac{1}{\sqrt{C_d}}$. [2], [5]

3.2. Experimental Apparatus and Environment

The acquisition of high-fidelity kinematic data necessitates a carefully structured testing environment. This section describes the physical arrangement of the drop zone, the optical instrumentation used for trajectory recording, and the rationale behind the chosen acceleration height to guarantee that steady-state descent conditions are met prior to measurement.

3.2.1. Drop zone and spatial calibration

Measurements were performed along a vertical, uniform reference surface, serving as a contrast background. The measurement window was delimited by three horizontal high-contrast fiducial markers spaced at exactly 0.50 m with uncertainty of $\delta h = \pm 0.01$ m, resulting from the limitations of the accuracy of reading and determining the positions of the markers in the image. This arrangement determined a total vertical measurement range of 1.0 m and two independent length intervals of 0.50 m: section A–B and B–C. Dividing the window into two intervals made it possible to verify the constancy of speed (the terminal velocity criterion v_t) by comparing the average speeds calculated separately for A–B and B–C.

3.2.2. Recording instrumentation and test stand geometry

The motion trajectory was recorded using a Nikon D850 camera with a Nikon AF-S NIKKOR 70–200 mm f/2.8G ED VR II lens mounted at a distance of 4.5m from the calibration wall (fig. 3.3). A long focal length (70–200 mm) combined with a 4.5m stand-off distance was utilized to approximate an orthographic projection, thereby minimizing perspective distortion and parallax error at the boundaries of the measurement window. The optical axis was set strictly perpendicular to the wall plane and directed towards the centre of the measurement window (centre marker) to minimise geometric scale deviations between the upper and lower parts of the image.

The recording was carried out at a frequency of 60 fps, which corresponds to a time resolution of $\delta t \approx 16.7$ ms per frame. This sampling frequency allowed the transit times through 0.50 m sections to be determined by counting frames with relatively low time quantisation compared to the typical transit times in the experiment.

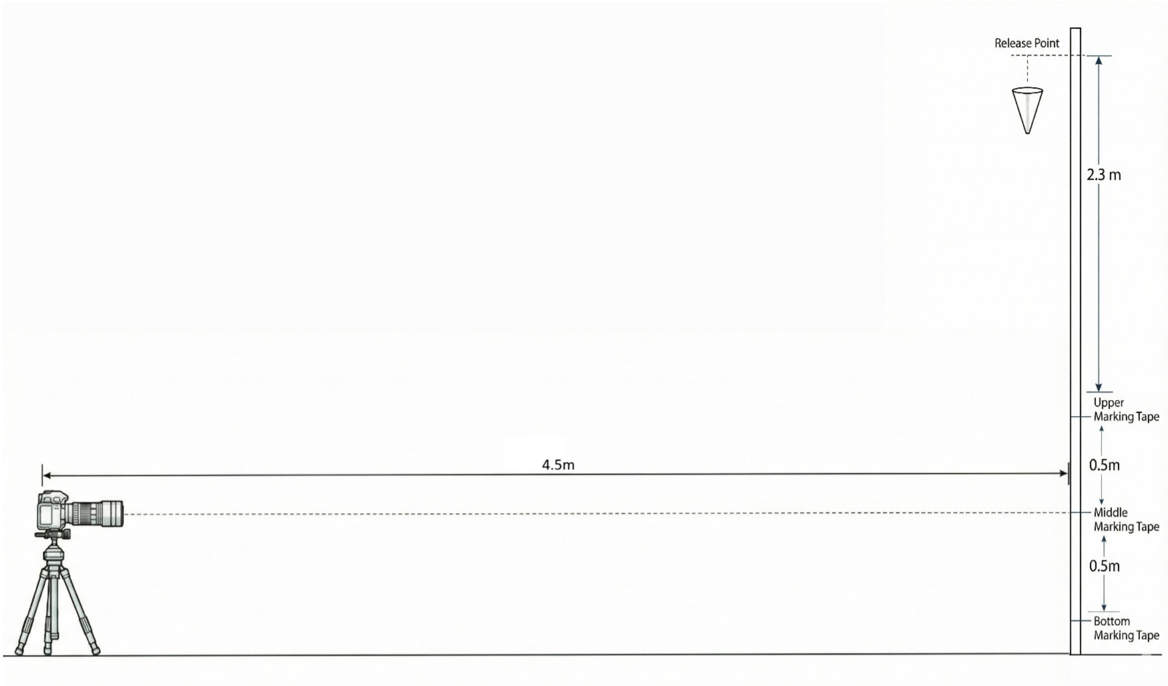


Figure 3.3: *Diagram of the test stand geometry: distance between the camera and the wall, position of calibration markers, and height of the drop point relative to the measurement window (generated with help of AI).*

3.2.3. Drop geometry and acceleration zone

The cones were released in a tip-down configuration from a height of approximately 3.3 m above the lowest marker. This means that before entering the measurement window (above the first marker), the cone travelled an acceleration distance of approximately 2.3 m. This distance was important in order to achieve a sufficiently high percentage of the terminal velocity: the aim was to ensure that at the start of recording in the measurement window, the velocity of the cone was at least equal to 95% of its terminal velocity v_t.

For aerodynamic drag proportional to the square of velocity, solving the equation of motion leads to a relationship describing how fast the object approaches v_t. In particular, the relative velocity after travelling a vertical distance y can be expressed as;

$$\frac{v(y)}{v_t} = \sqrt{1 - \exp\left(-\frac{2gy}{v_t^2}\right)}. \tag{3.2}$$

For $y = 2.3\,\mathrm{m}$ and experimentally determined values of v_t, it was found that the cones reached the following fraction of the terminal velocity even before entering the measurement window:

- 30° ($v_t \approx 3.71\,\mathrm{m/s}$): $v(y) \approx 98.1\%\, v_t$,
- 60° ($v_t \approx 3.36\,\mathrm{m/s}$): $v(y) \approx 99.1\%\, v_t$,
- 90° ($v_t \approx 2.90\,\mathrm{m/s}$): $v(y) \approx 99.8\%\, v_t$,
- 120° ($v_t \approx 2.75\,\mathrm{m/s}$): $v(y) > 99.9\%\, v_t$,
- 150° ($v_t \approx 2.52\,\mathrm{m/s}$): $v(y) > 99.9\%\, v_t$.

These results confirm that the acceleration zone of 2.3 m was sufficient to meet the required 95% of terminal velocity (v_t) and the assumption of steady motion in the measurement window with a high degree of accuracy.

3.2.4. Environmental conditions

The local fluid density was calculated as $\rho = 1.1937\,\mathrm{kg/m^3}$ using the ideal gas law parametrized with ambient atmospheric pressure, temperature, and relative humidity recorded contemporaneously with the drop tests and converted using an online calculator. [12] Adopting a consistent value for ρ was necessary because it appears directly in the drag equation and in the relationship for the terminal velocity, and therefore affects the calculated values of C_d.

3.3. Measurement Protocol and Data Acquisition

Ensuring repeatability and aerodynamic stability during the free-fall tests is important for accurate drag characterization. This section delineates the drop procedure, the stability criteria applied during sample selection, and the frame-by-frame video analysis techniques utilized to extract the terminal velocity (v_t) from the raw footage.

3.3.1. Drop procedure

Specimens were released manually from rest in a vertex-down orientation. The 2.3 m acceleration zone served a dual aerodynamic purpose: allowing the specimens to asymptotically approach ≥ 98% of their terminal velocity, and providing sufficient time for aerodynamic restoring moments to damp out any initial angular perturbations induced by the manual release. Regardless of this assumption, the fulfilment of the steady-state condition was verified directly on the basis of video data by comparing two consecutive velocity sections (3.2.1).

3.3.2. Sampling: *Sequential Inclusion* protocol

Due to the possibility of unstable behaviour a strict kinematic stability filter was implemented to ensure valid application of the steady-state drag equation (eq. (2.1)). The first six trials with trajectories exhibiting purely vertical, quasi-steady descent were accepted for further analysis. Trials exhibiting dynamic instabilities (e.g., vortex-induced flutter, tumbling, or lateral drift) were systematically excluded, as these behaviours indicate a transition to an unstable aerodynamic regime in which the assumption of equilibrium of forces along the descent axis (and thus a unique value of v_t and C_d) cannot be calculated. Consequently, the rejected tests were not used to determine v_t or to further investigate the relationship $C_d(\alpha)$.[4], [9]

3.3.3. Video analysis and data extraction

The recorded video material was imported into Adobe Premiere Elements 2023 and analysed frame by frame. The apex of the cone was taken as the tracking point, whose position was read relative to the scaled background. Recording at a frequency of 60 fps provided a temporal resolution of approximately

0.0167 s per frame. For each transition between adjacent markers with a known distance of $y = 0.50\,\mathrm{m}$, the number of frames n was determined, followed by the transit time.

$$t = \frac{n}{60}. \tag{3.3}$$

The velocity on a given section was calculated as

$$v = \frac{y}{t}. \tag{3.4}$$

In order to verify that the cone had reached a stable terminal velocity v_t within the measurement window, the descent speed was determined independently for two consecutive vertical sections of 0.50 m (intervals AB and BC). The criterion for stability was defined based on the temporal resolution of the recording equipment: the transit times t_{AB} and t_{BC} were required to agree within the limits of frame discretization error ($\delta t \leq 1/60\,\mathrm{s}$, corresponding to a maximum deviation of one frame). In the collected dataset, no trials exceeded this threshold, confirming that the motion was uniform and that terminal velocity had been successfully attained prior to the measurement zone.[4]

For every shape, six accepted velocity values were compiled, the average v_t was calculated, and input data was prepared for further determination of C_d and statistical analysis. The final aggregation of results and statistical processing were performed in Microsoft Excel.

3.4. Error Analysis and Uncertainty Propagation

A quantification of measurement errors is needed to evaluate the confidence intervals of the experimentally derived drag coefficients. This section systematically identifies the primary sources of geometric and instrumental uncertainties, details their propagation through the governing aerodynamic equations, and discusses the statistical treatment of data variability across the experimental trials.

3.4.1. Geometric and Instrumental Uncertainties

In order to ensure comparability of results between different geometries and to enable subsequent propagation of uncertainty to derived quantities (subsection 3.4.3), all primary sources of measurement error related to sample geometry and recording equipment resolution were identified and quantified. This subsection summarises the directly measured uncertainties; the uncertainty of the angle α is discussed separately in subsection 3.4.2.

The basic geometric parameter affecting aerodynamic quantities is the base radius r, which determines the reference surface area A (for a downward-pointing tip). The construction and measurement of the radius were subject to a tolerance:

$$\delta r = \pm 0.1\,\mathrm{cm}.$$

The uncertainty δr translates directly into the uncertainty δA (since $A = \pi r^2$), and thus affects the subsequently determined C_d and v_t. The value of A together with its uncertainty was determined at the test sample preparation stage and treated as a controlled variable.

The measurement section was determined using markers on the background wall, spaced evenly at $y = 0.50\,\mathrm{m}$. The accuracy of determining this distance was limited by the tolerance of the marker placement (as well as their width and blurring of edges in the video material). In the analysis, a conservative uncertainty in distance reading was assumed:

$$\delta y \approx \pm 0.01\,\mathrm{m}.$$

This uncertainty directly affects the velocity determined by the kinematic method $v = \frac{y}{t}$.

The transit time was determined by counting the number of frames n needed to cover the distance y in material recorded at a frequency of 60 fps. Due to the discrete nature of time sampling, the minimum unit of time corresponded to one frame:

$$t_{\text{frame}} = \frac{1}{60}\,\text{s} \approx 0.0167\,\text{s}.$$

Consequently, the fundamental uncertainty in determining the crossing time, resulting from the identification of the marker's entry/exit frame, was assumed to be:

$$\delta t = \pm\frac{1}{60}\,\text{s}.$$

This is the dominant source of instrumental uncertainty at short transit times and is the main limitation on the velocity resolution achievable with the frame-by-frame method.

The uncertainties listed above (δr, δy and δt) directly concern the measured input quantities and will be used in section 3.4.3 to determine the uncertainties of velocity and C_d using error propagation methods [13].

3.4.2. Uncertainty in angle measurement

The accuracy of the cone construction was governed by the precision of the geometric layout, specifically the measurement of the base radius ($r = 6.0 \pm 0.1\,\text{cm}$) and the corresponding slant height (R), both subject to a manufacturing tolerance of $\pm 0.1\,\text{cm}$. The uncertainty in the apex angle α was derived using standard error propagation applied to the geometric relationship $\sin(\alpha/2) = r/R$. Analysis of this relationship reveals a non-linear sensitivity: as the cone becomes flatter (approaching 180°), small deviations in the measured slant height result in increasingly large deviations in the resulting angle. Consequently, the structural uncertainty varied across the test set: while the acute 30° cone maintained a high precision of ±0.5°, the error margin expanded progressively for wider geometries, up to uncertainty of ±9.9° for the obtuse 150° cone.

Cone Angle [deg]	**Uncertainty [± deg]**
30	0.53
60	1.23
90	2.34
120	4.38
150	9.91

Table 3.1: *Structural uncertainty of the cone apex angle (α) resulting from manufacturing tolerances ($\delta r = \delta R = \pm 0.1\,cm$).*

3.4.3. Propagation of Uncertainty to Derived Quantities

The values reported in the results section, i.e., the terminal velocity v_t and the drag coefficient C_d, are determined on the basis of directly measured values: distance y, flight time t, base radius r (i.e., reference area A), mass m, and air density ρ. Consequently, the uncertainty δC_d is a combination of four main contributions: the uncertainty of mass (m), the reference area (A), the air density (ρ), and the square of velocity (v^2). For independent sources of uncertainty, the standard Root-Sum-Square (RSS) method was used. [13], [14]

1) Uncertainty of velocity δv. The velocity on the calibration section was determined from the relationship;

$$v = \frac{y}{t}.$$

For the quotient of two measured quantities, the relative uncertainty of velocity is equal to:

$$\frac{\delta v}{v} = \sqrt{\left(\frac{\delta y}{y}\right)^2 + \left(\frac{\delta t}{t}\right)^2}. \tag{3.5}$$

Since δt is limited by the time resolution of the camera (one frame), the time component dominates especially for shorter flight times (faster cones). In fact, the relative velocity error was approximately 12.5% for the fastest cone (30°) and approximately 8.6% for the slowest (150°).

2) Uncertainty of the reference area (δA). Since the area depends on the square of the radius, the relative uncertainty is derived using the power rule for error propagation [13]:

$$\frac{\delta A}{A} = 2\frac{\delta r}{r}. \tag{3.6}$$

For $\delta r = 0.1\,\text{cm}$ and $r = 6.0\,\text{cm}$, a constant relative contribution is obtained

$$\frac{\delta A}{A} \approx 3.3\%.$$

3) Uncertainty of mass (δm). The mass of all samples was adjusted to $m = 0.005\,\text{kg}$, and the uncertainty was $\delta m = 0.0001\,\text{kg}$, which gives a constant relative contribution:

$$\frac{\delta m}{m} = \frac{0.0001}{0.005} = 0.02. \tag{3.7}$$

This yields an uncertainty of 2.0% for each case.

4) Uncertainty of air density ($\delta \rho$). Since the ambient conditions were monitored but not continuously logged during the trials, a conservative relative uncertainty of 1% was assumed for the air density ρ. This estimate accounts for potential minor fluctuations in temperature and atmospheric pressure.

5) Total uncertainty of the drag coefficient (δC_d). The drag coefficient is derived from the balance of forces at terminal velocity:

$$C_d = \frac{2mg}{\rho A v^2} \tag{3.8}$$

Applying the error propagation rule for products and quotients, the relative uncertainty of C_d is given by the quadratic sum of the relative uncertainties of its components:

$$\frac{\delta C_d}{C_d} = \sqrt{\left(\frac{\delta m}{m}\right)^2 + \left(\frac{\delta A}{A}\right)^2 + \left(\frac{\delta \rho}{\rho}\right)^2 + \left(2\frac{\delta v}{v}\right)^2} \tag{3.9}$$

Because velocity governs the drag force quadratically, its relative uncertainty dominates the propagation via a sensitivity coefficient of 2 (i.e. $2\delta v / v$). Based on the experimental data, the total relative uncertainty of the drag coefficient was found to range from approximately 17.7% for the slowest cone (150°) to 25.4% for the fastest cone (30°).

Table 3.2 presents the relative error values used in the rest of the study.

α [°]	$\delta v/v$ [%]	$\delta A/A$ [%]	$\delta m/m$ [%]	$\delta\rho/\rho$ [%]	$\delta C_d/C_d$ [%]
30	12.5	3.3	2.0	1.0	25.4
60	11.4	3.3	2.0	1.0	23.1
90	9.9	3.3	2.0	1.0	20.2
120	9.4	3.3	2.0	1.0	19.2
150	8.6	3.3	2.0	1.0	17.7

Table 3.2: *Table of relative propagated uncertainties.*

3.4.4. Statistical Treatment of Variability

The raw data obtained from the frame analysis contain both random dispersion resulting from minor aerodynamic fluctuations (e.g., minimal trajectory deviations) and instrumental uncertainty resulting from measurement resolution. In order to ensure the reliability of the estimation, for each angle α the values averaged on the basis of $N = 6$ trials accepted in accordance with the selection protocol (section 3.3) were reported.

For each geometry, the final result was determined as the arithmetic mean $\bar{x}$ from $N = 6$ valid trials. Averaging reduces the impact of random, minor instabilities that may introduce scatter in individual measurements but do not change the steady-state settling regime.

The precision of the mean estimate is described by the Standard Error (SE) of the mean, defined as [15]:

$$SE = \frac{\sigma}{\sqrt{N}}, \tag{3.10}$$

where σ is the standard deviation of the measurement series and N is the number of samples. The Standard Error (SE) of the velocity measurements was constrained to $SE_v \approx \pm 0.04\,\mathrm{m/s}$, demonstrating a high degree of intra-sample repeatability across the filtered dataset.

It is important to distinguish between two types of uncertainty:

- **Instrumental uncertainty** (Subsection 3.4.3), resulting from the limitations of the method (predominantly: camera time discretisation and determination of δt).
- **Statistical uncertainty** of the mean, described by SE, which was approximately equal to $SE_v \approx \pm 0.04\,\mathrm{m/s}$.

To avoid overstating the declared accuracy and maintain conservative reporting rigour, the error bars presented in the result graphs reflect the greater instrumental uncertainty from 3.4.3 rather than the smaller standard error of the mean. This approach prevents overinterpretation of the precision of the results and provides a cautious assessment of the actual accuracy of the experiment. [13], [14]

4
Experimental Results

This chapter presents the empirical data obtained from the free-fall experiments outlined in chapter 3. First, section 4.1 details the measured terminal velocities for each conical geometry, demonstrating the kinematic impact of the shape variations. Building upon these results, section 4.2 derives the corresponding drag coefficients, quantifying the aerodynamic penalty associated with increasing apex angles. The data presented herein serves as the foundation for the subsequent analytical modelling and numerical validation.

4.1. Measurements of Terminal Velocity

This section presents the results of measurements of the terminal velocity v_t for five cone shapes with different apex angles α. Terminal velocities (v_t) were extracted from the stabilized trajectory phase for each geometry. As detailed in Table 4.1, a strict monotonic decrease in terminal velocity v_t was observed as the apex angle α increased. For clarity, the complete set of raw tracking data (frame numbers and transition times for each sample and each cone) is available in Appendix chapter A.

Table 4.1: *Table of average terminal velocity v_t with uncertainties (values rounded to 3 decimal places).*

α [°]	$\delta\alpha$ [°]	v_t [m/s]	$\delta v/v$ [%]	δv [m/s]
30	0.529	3.711	12.5	0.466
60	1.233	3.364	11.4	0.384
90	2.339	2.903	9.9	0.288
120	4.376	2.748	9.4	0.258
150	9.910	2.517	8.6	0.218

The most slender cone (30°) reached a terminal velocity of approximately $v_t \approx 3.71\,\mathrm{m/s}$, while the bluffest geometry (150°) exhibited a 32% reduction in descent rate, stabilizing at $v_t \approx 2.52\ m/s$. Since the mass m and reference area A were constant for all samples (3.1), this difference indicates a significant increase in aerodynamic drag resulting solely from the change in geometry, i.e. from the increase in C_d for larger angles α.

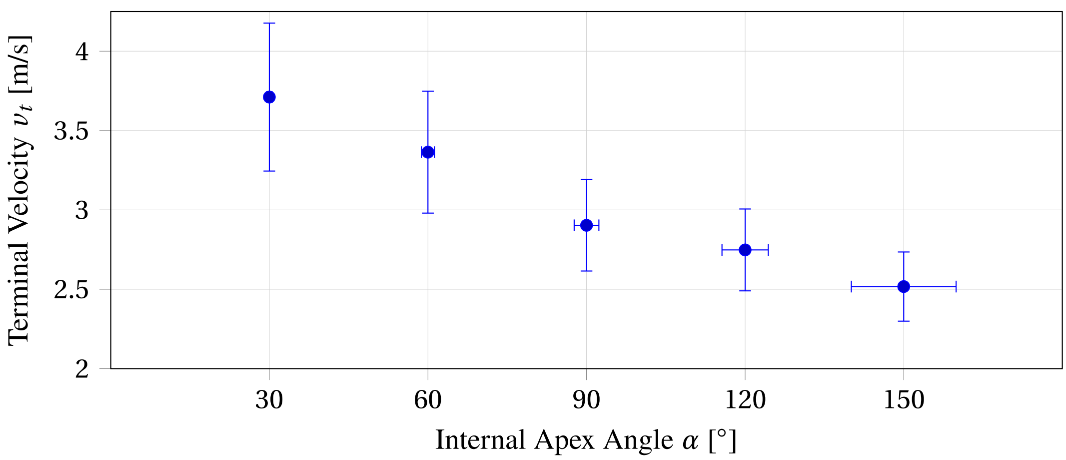


Figure 4.1: *Experimental terminal velocities (v_t) and associated instrumental uncertainties, demonstrating an inverse monotonic relationship with apex angle α.*

4.2. Drag Coefficient Determination

The drag coefficient C_d was determined based on the terminal velocity v_t obtained experimentally in section 4.1. Drag coefficients (C_d) were computed by isolating the coefficient in the steady-state equilibrium condition established in eq. (2.3). The C_d values were calculated for each geometry using the constant parameters m, ρ, and A (section 3.1) alongside the averaged velocities v_t. The uncertainties for C_d were determined in accordance with the error propagation framework described in section 3.4.3.

Table 4.2: *Summary of experimental mean values C_d with uncertainties (values rounded to 3 decimal places).*

α [°]	$\delta\alpha$ [°]	C_d [-]	$\delta C_d / C_d$ [%]	δC_d [-]
30	0.529	0.528	25.4	0.134
60	1.233	0.642	23.1	0.148
90	2.339	0.862	20.2	0.174
120	4.376	0.962	19.2	0.184
150	9.910	1.147	17.7	0.203

A clear, non-linear dependence of C_d on the angle α was observed: the drag coefficient more than doubles, escalating from $C_d \approx 0.528$ for the slender 30° profile to $C_d \approx 1.147$ for the 150° geometry. This $> 100\%$ amplification empirically isolates the severe aerodynamic penalty of bluff geometries, validating that form drag scales aggressively with the internal apex angle even when mass and frontal area are strictly controlled.

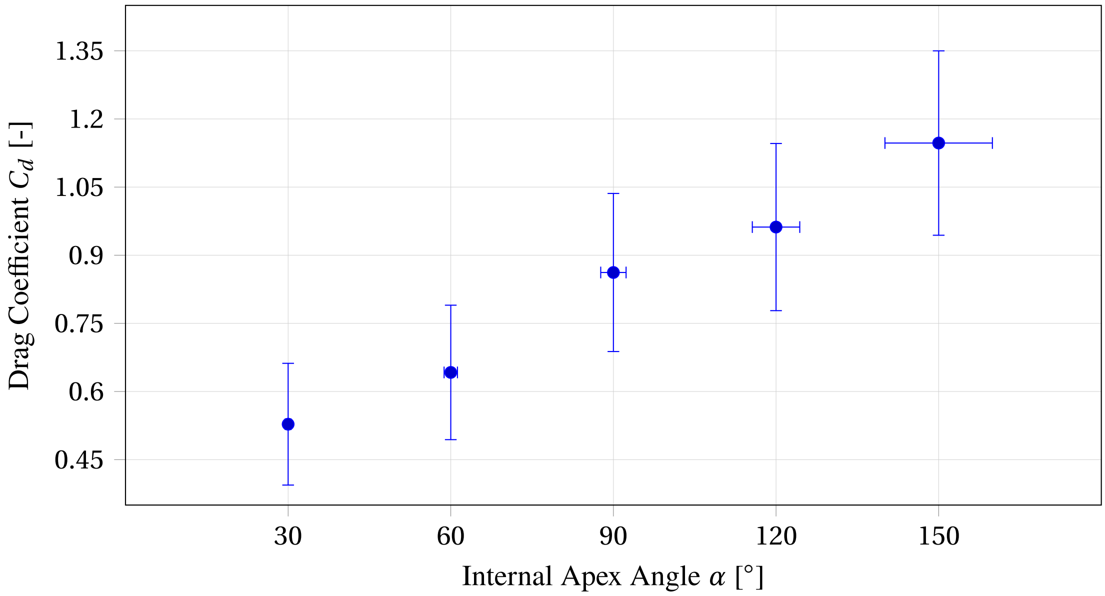


Figure 4.2: *Dependence of the experimental drag coefficient C_d on the internal apex angle α with error bars ($\delta\alpha$ and δC_d). The error bars were determined in accordance with section 3.4.3.*

5
Numerical Validation (CFD)

The primary objective of this chapter is to complement the experimental drop tests with a rigorous numerical analysis. The Computational Fluid Dynamics (CFD) study serves a dual purpose: first, to validate the empirically derived drag coefficients against a controlled theoretical model; and second, to enrich the understanding of the aerodynamic mechanisms by visualizing flow field characteristics, such as wake topology and surface pressure distribution, that were experimentally inaccessible. To ensure a direct one-to-one comparison with the physical experiments, the digital geometries were constructed using the exact geometric parameters defined in Chapter 3, maintaining a constant base radius ($r = 6.0\,\mathrm{cm}$) and fixed surface area (A) across all investigated half-angles (α), thereby guaranteeing high geometric fidelity to the physical test specimens.

5.1. Computational Domain and Solver Settings

The computational domains were discretized and solved using the OpenFOAM finite-volume framework. The steady-state incompressible Reynolds-Averaged Navier–Stokes (RANS) equations were resolved via the Semi-Implicit Method for Pressure Linked Equations (SIMPLE) algorithm. The choice of a steady-state solution was justified by the purpose of the work: the experiment and analysis focus on the terminal velocity v_t, which corresponds to the state of equilibrium of forces ($F_d = F_g$). In this state, the average (time-averaged) value of the drag force is of interest, and thus also the average C_d coefficient; thus, the steady-state solution is sufficient and computationally more efficient than full unsteady modelling. The boundary conditions were defined in such a way as to reproduce the conditions of free fall in an open environment (without wind tunnel effects) as accurately as possible. The following boundary conditions were applied [16], [17], [18]:

- ***Velocity Inlet.*** A Dirichlet boundary condition was applied at the inlet, where the velocity vector is explicitly prescribed as uniform with a magnitude corresponding to the experimental terminal velocity v_t for a given angle α. Specifying a fixed velocity at the inflow while leaving pressure to be extrapolated from the interior is a standard requirement for ensuring a well-posed problem in incompressible subsonic flow simulations.

- ***Pressure Outlet.*** At the downstream boundary, a Pressure Outlet condition was imposed with a fixed kinematic gauge pressure $\frac{p}{\rho} = 0$. This applies a Dirichlet condition for pressure while implicitly enforcing a Neumann condition ($\partial\phi/\partial n = 0$) on the velocity field. This configuration allows flow structures and wake turbulence to exit the domain without artificial reflection, provided the boundary is placed sufficiently far downstream.

- **Far-field (*Slip Wall*).** The outer radial boundaries of the domain were modeled using a slip condition (symmetry). Mathematically, this enforces zero normal velocity ($u_n = 0$) and zero normal gradients for all scalar variables. This approach approximates far-field free-stream conditions, effectively eliminating the artificial blockage effects and wall boundary layers that would arise with standard no-slip walls in a finite wind-tunnel domain.

- **Cone surface (*No-slip Wall*).** A no-slip condition was strictly enforced on the surface of the cone ($u_{\text{fluid}} = u_{\text{wall}} = 0$). This is necessary to correctly resolve the large velocity gradients near the solid boundary, accurately capturing the boundary layer development and the viscous skin friction contribution to the total drag coefficient C_d.

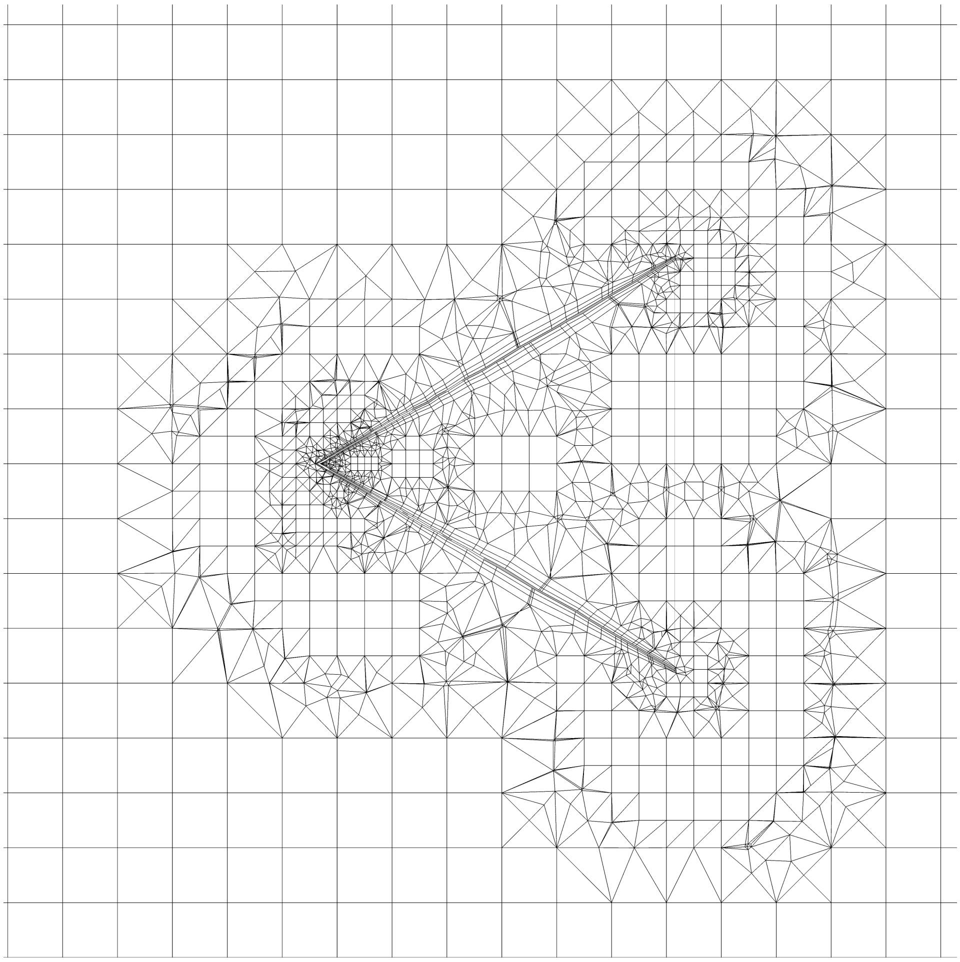


Figure 5.1: *Longitudinal cross-section of the computational mesh. The visualization highlights the refinement regions in the wake zone and the inflation layers adjacent to the cone surface, ensuring accurate resolution of the boundary layer.*

5.2. Turbulence Modelling Strategy

All simulations were performed using the averaged Navier-Stokes equations (RANS), which required the use of a turbulence closure. The selection of turbulence models was deliberate: the base model was chosen to reproduce the flow separation and pressure distribution for blunt cones as accurately as possible, while for the most slender geometry, a sensitivity comparison was introduced to better reflect the actual (non-ideal) experimental conditions.

5.2.1. Base model: Menter's k-ω SST

The k–ω Shear Stress Transport (SST) model was adopted as the basic turbulence closure, used in all cases as the reference solution. This model combines the advantages of two closure families through a switching function: near the wall, it uses the k-ω form (which favours the correct mapping of the boundary layer), while in the free flow region, it switches to behaviour similar to k-ϵ (reducing sensitivity to conditions in the far field). This approach is particularly advantageous in external aerodynamics, where the correct capture of boundary layer development and separation moment in the presence of an unfavourable pressure gradient is critical for determining drag [19], [20].

The selection of the turbulence model was driven by two essential flow characteristics: the specific Reynolds number regime ($Re \approx 10^4 - 3 \times 10^4$) and the dominance of pressure drag in the blunt geometries ($60° - 150°$). In these cases, the aerodynamic forces are governed by flow separation at the base edge and the resulting wake topology. The SST model was therefore adopted as the baseline model because it allows for stable determination of averaged velocity and pressure fields in the presence of separation and provides a consistent comparative methodology between all α angles.

5.2.2. Sensitivity analysis: the case of a 30° cone (the k-ϵ model)

For the highly slender 30° geometry, drag prediction is exceptionally sensitive to the exact locus of boundary layer separation. Because steady-state RANS formulations inherently struggle to resolve unforced, adverse-pressure-gradient induced separation on slender profiles, a sensitivity analysis was conducted utilizing the realizable k–ϵ turbulence model (k–ϵ) model. This provides a comparative bound on the accuracy of turbulence closure, highlighting how differing formulations of eddy viscosity impact the predicted wake topology, independent of structural idealizations. The use of k–ϵ in this single case was therefore of a control nature and was used solely to assess the impact of the choice of turbulence closure on the consistency of the C_d prediction with the experimental data, a detailed interpretation of this and its consequences has been moved to the Chapter 6. [21], [22]

5.3. Numerical Simulation Outputs

This section presents the quantitative and qualitative outcomes of the computational analysis. It begins by establishing the numerical convergence of the simulation domains, followed by a detailed compilation of the computed drag coefficients for each geometry. Finally, flow field visualizations are provided to elucidate the physical mechanisms—such as wake topology and stagnation zones—that drive the observed aerodynamic behaviour.

5.3.1. Numerical Stability and Convergence

Numerical convergence was established by monitoring the iterative history of the drag coefficient alongside the domain residuals. The steady-state solution was deemed converged when the computed force coefficients reached an iterative asymptote (fluctuations $< 0.1\%$).

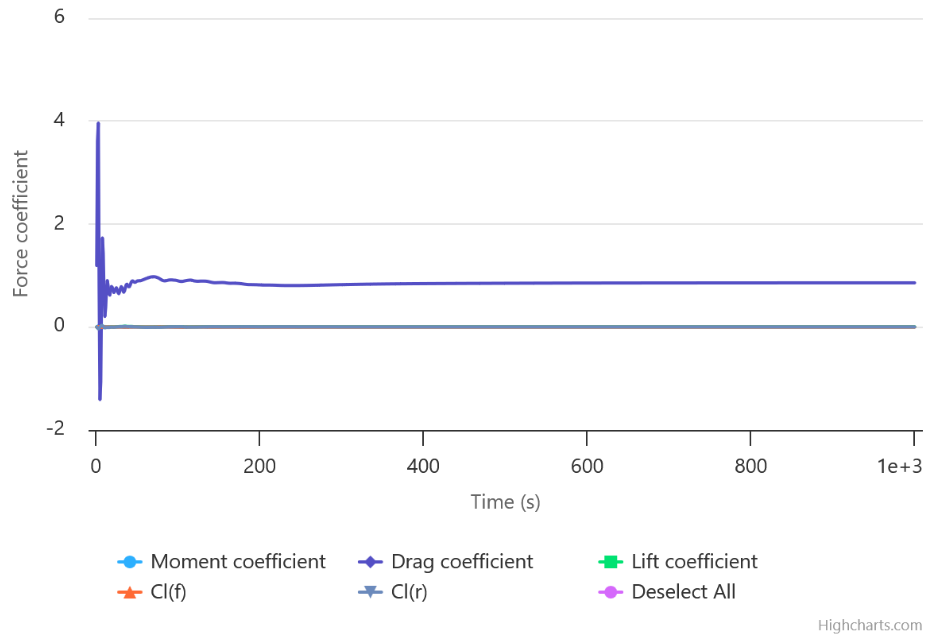


Figure 5.2: *Convergence history for cone with $\alpha = 90°$ of the drag coefficient (C_d) versus iteration number. The flattening of the curve demonstrates that the solution has reached a stable equilibrium.*

5.3.2. Drag Coefficients obtained using Computational Fluid Dynamics

The aim of this subsection is to present the raw numerical results obtained from the CFD simulations. The calculations were executed utilizing the domain and solver configurations outlined in section 5.1. The computed C_d values, extracted after 1000 iterations for each geometry, are summarized in table 5.1. These isolated numerical outputs serve as a theoretical baseline for the subsequent comparative analysis with the experimental data in chapter 6.

α [°]	C_d (SST) [-]	C_d (k–ϵ) [-]
30	0.384	0.480
60	0.642	—
90	0.857	—
120	1.040	—
150	1.167	—

Table 5.1: *The C_d values obtained numerically after 1000 iterations (RANS). For the 30° cone, both the baseline SST result and the k–ϵ sensitivity analysis result are provided. Values are rounded to 3 decimal places.*

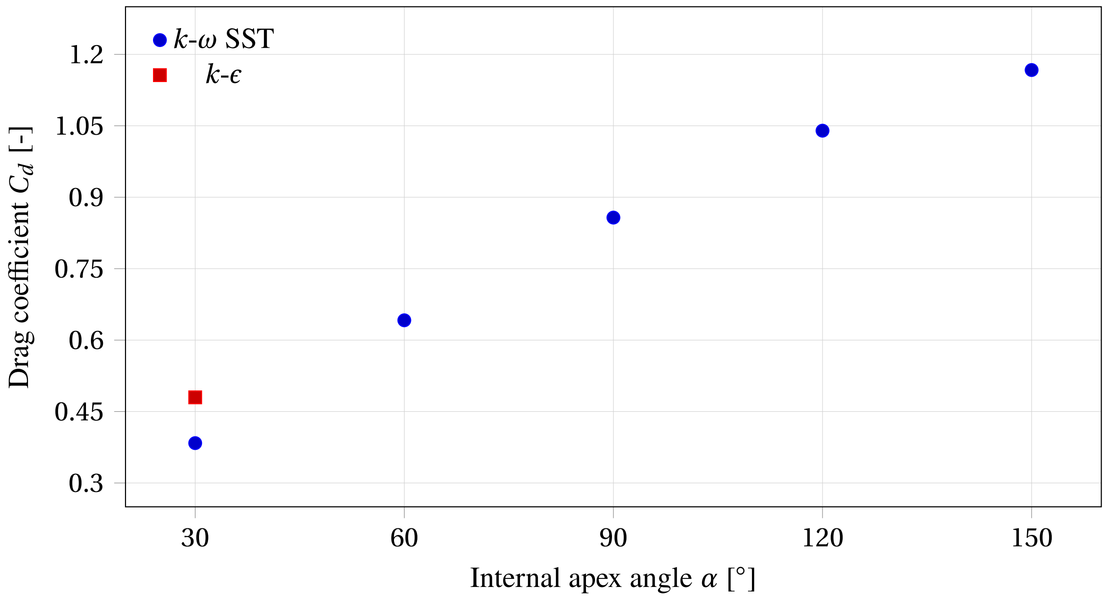


Figure 5.3: *Dependence of the numerical drag coefficient C_d on the apex angle α. Blue dots: SST model. Red dot: result of k–ϵ sensitivity analysis for a 30° cone.*

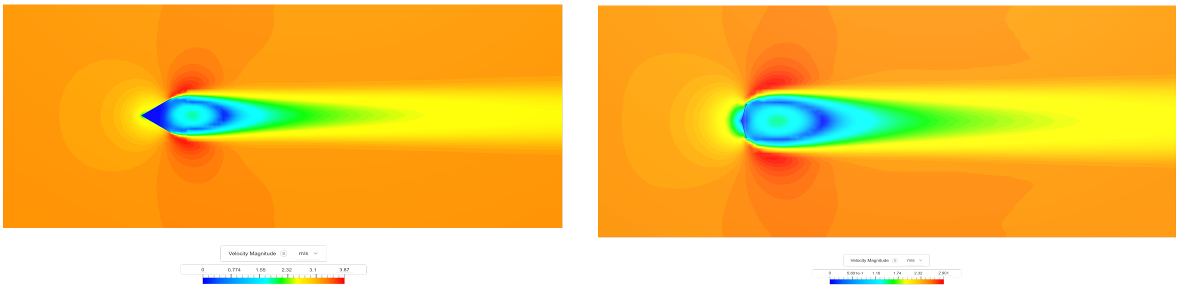
A clear, non-linear increase in C_d was observed with increasing angle α: simulations predict a rapid increase in drag for increasingly 'blunt' cones, which is consistent with the expected increase in the proportion of pressure drag resulting from separation and wake development. The discrepancy is particularly notable in the case of 30°: the result SST ($C_d \approx 0.384$) differs significantly from the result k–ϵ ($C_d \approx 0.480$). This difference is the direct motivation for a detailed analysis of the causes of the deviation in Chapter 6.

5.3.3. Flow Visualization

In order to correlate the quantitative C_d results (section 5.3.2) with the underlying flow physics, a qualitative analysis of the flow fields generated by the CFD simulations was performed. Visualizations were utilized to identify key aerodynamic structures, specifically:

- the scale and topology of the wake region associated with flow separation,
- the surface pressure distribution, including the high-pressure stagnation zone near the apex.

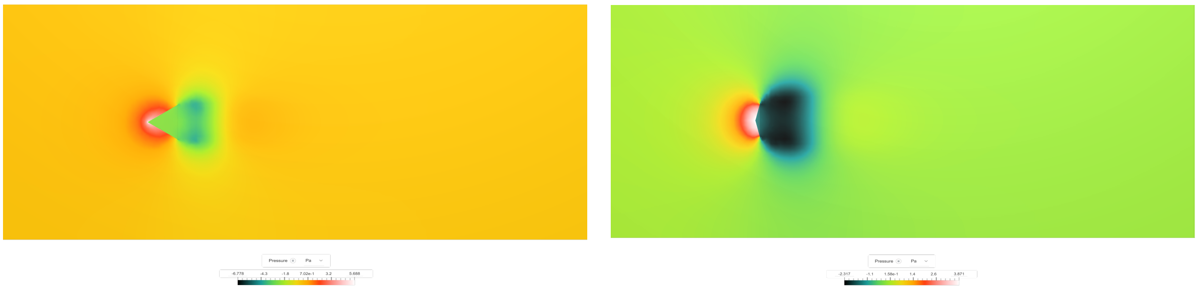
Velocity field. Longitudinal cut-planes of the velocity magnitude vividly map the near-wake recirculation regions. As compared in fig. 5.4, the bluff 150° shell generates a massive, persistent wake deficit, whereas the 60° cone exhibits a narrowly confined shear layer.



(a) $\alpha = 60°$ **(b)** $\alpha = 150°$

Figure 5.4: *Comparison of velocity profile contours (longitudinal section) for a* 60° *cone (left) and a* 150° *cone (right). The visualisation highlights the differences in the size of the wake and the separation zone.*

Pressure distribution and stagnation zone. In addition, pressure maps were generated, enabling the identification of high and low pressure areas on the surface and in the vicinity of the object. A characteristic feature is the high-pressure zone in the vicinity of the apex (stagnation zone) where the stream is strongly decelerated to a velocity close to zero and the static pressure reaches a local maximum. Pressure maps for 60° and 150° cones are shown in fig. 5.5.



(a) $\alpha = 60°$ **(b)** $\alpha = 150°$

Figure 5.5: *Pressure distribution maps for* 60° *and* 150° *cones, showing the stagnation zone (high pressure) near the apex and areas of reduced pressure behind the object.*

All collected visualisations are compiled in Appendix B. Furthermore, the complete raw simulation data (output files) for all cases can be downloaded using the link provided in Appendix D.

6
Discussion and Analysis

This analysis synthesizes the experimental measurements and CFD predictions to evaluate the geometric dependence of the drag coefficient $C_d(\alpha)$. By comparing the datasets, this chapter identifies the physical divergence caused by aeroelasticity in slender cones and validates a semi-empirical model to correct the classical Newtonian approximation.

The discussion is structured to first evaluate the blunt geometries ($60° - 150°$), where the flow separation point is geometrically fixed at the base edge, theoretically leading to more predictable aerodynamic behaviour. Subsequently, a sensitivity analysis is dedicated to the anomalous behaviour of the acute 30° cone. This specific case serves as a focal point for discussing the limitations of steady-state RANS turbulence modelling in regimes governed by complex reattachment and separation phenomena. Finally, the chapter synthesizes these findings to propose a corrected drag relationship that accounts for the observed geometric sensitivities.

6.1. Comparative Data Synthesis

Table 6.1 presents a summary of experimental and numerical results for C_d (model SST), together with experimental uncertainty C_d and relative prediction error CFD calculated as $\left(C_{d,\mathrm{CFD}} - C_{d,\mathrm{exp}}\right)/C_{d,\mathrm{exp}} \cdot$ 100%.

Table 6.1: *Comparison of experimental results and CFD for C_d (k-ω SST model).*

α [°]	$\delta\alpha$ [°]	$C_{d,\mathrm{exp}}$ [-]	$\delta C_d/C_d$ [%]	δC_d [-]	$C_{d,\mathrm{CFD}}$ [-]	Relative error [%]
30	0.529	0.528	25.4	0.134	0.384	−27.3
60	1.233	0.642	23.1	0.148	0.642	−0.1
90	2.339	0.862	20.2	0.174	0.857	−0.6
120	4.376	0.962	19.2	0.184	1.040	+8.1
150	9.910	1.147	17.7	0.203	1.167	+1.8

Figure 6.1 presents a direct comparison: blue points (experiment) with error bars and a red series (CFD prediction for k-ω SST). A consistent upward trend of C_d with increasing α is visible. Importantly, for all angles $\alpha \geq 60°$, the values of CFD are very close to the experimental values and, in fact, fall within the experimental uncertainty.

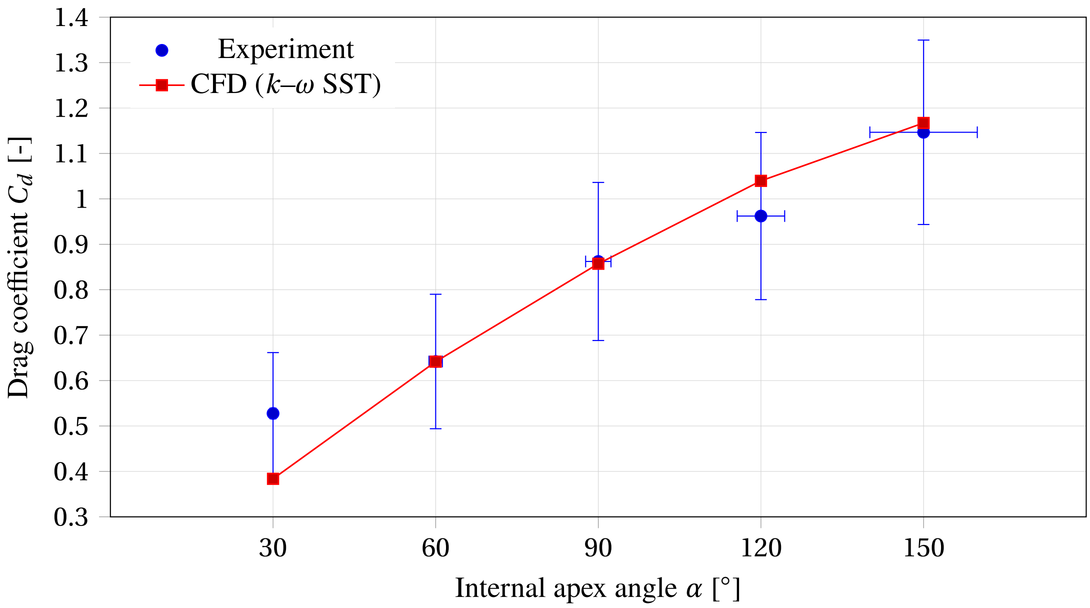


Figure 6.1: *Comparison of* $C_d(\alpha)$*: experiment (blue dots with error bars) and CFD predictions (red series,* k–ω *SST).*

6.1.1. Conformity assessment and conclusions for blunt geometries

The consistency between measurements and CFD predictions for cones with angles $\alpha \geq 60°$ is high both in quantitative terms (relative error) and qualitative terms (consistency of the trend and the position of the prediction relative to the experimental uncertainty intervals). From table 6.1 it follows that for $\alpha = 60°$ and $\alpha = 90°$ the differences between $C_{d,\text{CFD}}$ and $C_{d,\text{exp}}$ are less than 1%, which in practice means that the numerical values almost coincide with the experimental results. Such a small error is significant because the experimental uncertainty C_d (dominated by the uncertainty of velocity determination and its square in the drag equation) is of the order of 20%; the CFD prediction is therefore well within the uncertainty range, and the agreement is not accidental, but indicates high reliability of both methods.

The $\alpha = 150°$ cone exhibited a relative discrepancy of only 1.8%. This narrow margin is very important for a bluff geometry, demonstrating that the steady-state RANS formulation accurately resolves the dominant pressure drag resulting from the massive wake deficit. In this case, the deviation is positive ($C_{d,\text{CFD}} > C_{d,\text{exp}}$), which may result from the fact that the stationary RANS model with k-ω SST closure provides an averaged solution, which for strongly detached flows tends to over-predict wake stability (less blurred, more persistent recirculation area), and thus to moderately overestimate the pressure drag compared to the average of the actual, not completely steady descent. At the same time, the difference of $\sim 2\%$ remains substantially smaller than the uncertainty of the experiment, and therefore does not constitute grounds for questioning the validity of the model.[19], [20]

For a 120° cone, the relative deviation is greater, at around 8%, but the key observation is that the CFD prediction is still within the experimental uncertainty δC_d (for this geometry, $\delta C_d / C_d \approx 19\%$). This means that, in a statistical and metrological sense, there is no contradiction between the results: the difference is clear, but it is not relevant in light of the identified measurement limitations. In practical terms, the 120° case rather indicates that with increasing bluntness, the geometry becomes more sensitive to the details of the pressure distribution at the base and in the wake, while the experimental uncertainty is dominated by the velocity error (and thus by time discretisation). Consequently, even a moderate change in the predicted base pressure can lead to a noticeable shift in C_d, which, however, remains hidden in the wide error bars of the experiment. [13]

The observed high agreement for blunt geometries can be explained physically and numerically by two complementary arguments:

1. **Geometrically determined separation.** For bodies with sharp streamlined edges (in this case: the sharp edge of the cone base), the separation point is largely 'forced' by geometry rather than subtle viscous effects. The literature emphasises that for sharp, angular blunt bodies, the drag equation with a constant approximation C_d can be applied already for $Re \gtrsim 10^4$, because the key feature of the flow (separation) is related to the edge and depends less on Re. [8] This justifies why, within the scope of the experiment (Re of the order of 10^4), the values of C_d for blunt cones can be predicted stably using the same turbulence model and a similar computational configuration.

2. **Selection of turbulence closure appropriate for separation and adverse pressure gradient.** The Menter's k-ω SST model is widely used in external aerodynamics precisely because it improves the prediction of flows with separation and unfavourable pressure gradients, while maintaining good wall properties thanks to the k-ω part and lower sensitivity to conditions in the far field thanks to the switch towards behaviour k-ϵ. [19], [20] In the case of blunt cones ($60° - 150°$), it is precisely the pressure drag associated with separation and wake that determines C_d, so choosing SST as the base closure directly supports achieving good compliance.

The configuration of boundary conditions described in section 5.1 is also of significant importance. Imposing a velocity equal to the experimental v_t at the inlet ensures comparability of the Reynolds number and dynamic reference pressure between the experiment and the simulation, while the use of *slip* walls minimises the blocking effect, which could artificially overestimate the resistance in the limited domain. As a result, the difference between $C_{d,\mathrm{CFD}}$ and $C_{d,\mathrm{exp}}$ for blunt geometries is dominated by the actual (physical) limitations of averaged modelling, rather than by numerical artefacts resulting from the domain geometry.

In summary, the agreement for cones $\alpha \geq 60°$ strongly validates both the experimental method (selection of stable samples, determination of v_t and propagation of uncertainty) and the adopted numerical model (RANS + k-ω SST) in the field of blunt geometry with sharp edges. At the same time, the visible outlier $\alpha = 30°$ (error of -27%) indicates an additional mechanism not included in the idealised computational description and has therefore been separated for a separate analysis in the next subsection.

6.2. Analysis of Aerodynamic Divergence for the Slender Geometry ($\alpha = 30°$)

In contrast to obtuse geometries ($\alpha \geq 60°$), the 30° cone showed a notable disagreement. The experimental value of the drag coefficient was $C_{d,\mathrm{exp}} \approx 0.528$, while the baseline CFD simulation with SST closure gave $C_{d,\mathrm{CFD}} \approx 0.384$. This corresponds to an underestimation of drag by approximately -27%, which is in stark contrast to the errors of the order of $\lesssim 2\%$ observed for blunt cones.

6.2.1. Aeroelastic and Surface Deformation Effects

The observed -27% deviation is attributed to the structural and geometric divergence of the physical test specimen from the numerical idealization. The base model RANS with SST assumes a geometrically perfect, rigid and smooth shell, i.e. an object with a geometry that does not change over time and a "hydraulically smooth" surface. In this approach, the flow is solved as steady and averaged, and the resistance results from the averaged distribution of pressure and tangential stresses on the *stationary* geometry.

In the case of a physical cone of 30°, these assumptions are least satisfied for the following reasons:

- **Structural vulnerability (flutter/micro-vibrations).** The 30° cone has the largest lateral surface area in relation to its base area, which is why it was made of thinner and less rigid paper (0.2 mm) than the other samples. In free fall conditions, even slight micro-vibrations ('breathing' of the coating) can modulate the local angle of attack of surface fragments and disturb the boundary layer. Such non-stationary phenomena can lead to increased energy losses in the wake and to an effective increase in drag compared to the idealised rigid case. Similar modes of unstable descent (fluttering/chaotic) are known for thin solids falling in a viscous medium and result from the interaction of the vortex wake with the motion of the body. [23]
- **Effective reference field variability and wall layer disturbance.** Although nominally the reference area A is constant (determined by the base radius), micro-vibrations can temporarily increase the effective bluffness seen by the flow. The result can be a larger average wake width and lower pressure in the area behind the object, which directly increases pressure drag. In flows around blunt bodies, drag is dominated by the pressure contribution associated with the extensive low-pressure zone in the wake; even a small increase in this zone can significantly increase C_d. [8]
- **Surface imperfections.** The 30° cone was made of paper with a lower grammage than the other samples, which increases the likelihood of local crumpling and deformation. Such defects can accelerate the transition to more 'energetic' turbulence near the wall or thicken the wall layer, thereby changing the separation conditions and wake structure. This effect is not captured in the model, which assumes a perfectly smooth and non-deformable surface.

The resulting underestimation of C_d by the baseline model can therefore be interpreted as a consequence of an overly idealised description for the most sensitive (slender and susceptible) geometry: the stationary simulation 'sees' the cone as a stable object with lower losses and a smaller footprint than in the experiment.

6.2.2. Sensitivity analysis using the k-ϵ model

In order to verify whether the discrepancy could result from the nature of averaged turbulence modelling and the "smoothing" of non-stationarity by the stationary approach, a comparative simulation was performed for a 30° cone using the k-ϵ model (cf. section 5.2). In this simulation, $C_{d,\mathrm{CFD}} \approx 0.480$ was obtained, which reduces the difference compared to the experiment (the underestimation decreases from approx. −27% to approx. −9%).

The interpretation of this result is as follows: the SST model is optimised for correct separation prediction for smooth, rigid solids and often gives a 'cleaner' picture of wall flow and wake. [19], [20] However, in the case of a real 30° cone, there were additional sources of aerodynamic 'noise' (microvibrations and surface defects) which, on average, increase losses and can lead to greater effective drag. The k-ϵ model tends to have stronger diffusion (greater dissipative 'discipline') in the turbulence field, which can act as an approximate substitute for these imperfections: it effectively increases the predicted size of the wake and losses, raising C_d towards the experimental value.

(a) *$k-\omega$ SST (Baseline)* **(b)** *$k-\epsilon$ (Sensitivity Check)*

Figure 6.2: *Comparison of the velocity field and wake structure for the $\alpha = 30°$ cone. The standard $k-\epsilon$ model predicts a larger recirculation region, resulting in a drag coefficient closer to the experimental value measured for not "ideal" $30°$ cone.*

Concluding, the 30° case indicates that for very slender, not fully rigid conical shells, the idealised steady-state simulation RANS with SST may underestimate the drag because it does not model the flow-structure interaction or the resulting unsteadiness. At the same time, the fact that a simple sensitivity analysis (change in turbulence closure) shifts the prediction in the expected direction supports the hypothesis that the source of the discrepancy is the "real" instability and non-ideality of the sample, rather than an error in the experimental method itself.

6.3. Aerodynamic Regimes and Wake Topology

The aim of this section is to relate the observed C_d values to the physical flow mechanisms revealed by CFD visualisations. In particular, velocity profiles (5.4) and pressure maps (5.5) are compared to explain why drag increases with increasing α and why the Newtonian model discussed in chapter 2 is unable to correctly predict drag for the cones in question.

6.3.1. Aerodynamic Topology and Pressure Distribution

For these blunt geometries, the total aerodynamic resistance is dominated by form drag (pressure drag), which arises from the severe pressure differential between the windward face and the leeward base. As it is can be seen in Fig. 5.5, particularly for $\alpha = 150°$ cone, the flow cannot accelerate significantly around the steep flanks. Consequently, the region of high static pressure is not confined to the apex but extends across the entire projected frontal area, maintaining near-stagnation pressure. In contrast, for sharper cones (e.g., $\alpha = 60°$), the flow deflects more gradually, resulting in a geometrically smaller stagnation zone and a reduced pressure gradient on the inflow side.

The Leeward Wake Structure seen in Fig. 5.4 is intrinsically coupled with the pressure differential. The 150° cone generates a wide separation region and an extensive volume of low-velocity recirculation. Physically, a wider wake inhibits pressure recovery behind the object, maintaining a lower base pressure (stronger suction). The 60° cone, conversely, produces a narrower wake, allowing for better pressure recovery.

In conclusion the high drag coefficient observed for blunt cones ($C_d \approx 1.15$) is therefore the result of a dual mechanism: maximum compression on the front combined with maximum suction at the base. As the cone becomes sharper, both the stagnation footprint and the wake width decrease, reducing the pressure differential and thus the total drag. [8]

6.3.2. Physical Limitations of the Newtonian Approximation

The Newtonian model discussed in Chapter 2 is based on the 'shadow' hypothesis behind the object: the medium is treated as a stream of independent particles that interact with the solid only through local collisions, and the area behind the solid is in practice an 'empty' region with no active flow dynamics. This idealisation implies that what happens behind the object does not affect the distribution of pressure and drag in a way that results from actual fluid mechanics.

The CFD results (Fig. 5.4 and Fig. 5.5) clearly show that this assumption is incorrect. Instead of a passive ‘shadow’, a dynamic recirculation zone is created, in which air is sucked in, turned back and mixed in the wake behind the base. At the same time, a distinct point/area of stagnation is created on the inflow side, proving that the flow ‘reacts’ to the presence of the object even before it ‘hits’ the surface in the Newtonian sense of particles. These two elements — stagnation at the front and recirculation/low velocity at the rear — generate a real, spatially complex pressure distribution that cannot be reduced to a simple balance of particle momentum on the surface itself.

Consequently, the observed drag is not only a function of the local inflow geometry captured in the law of the type $\sin^2(\frac{\alpha}{2})$, but also considerably depends on the formation and size of the wake, i.e. a mechanism directly ignored in Newton’s model. This explains why the simple Newtonian relationship cannot correctly predict C_d for the cones studied in subsonic flow, and also justifies the introduction of a semi-empirical model in which there is a non-zero base component and a part dependent on the bluntness of the geometry (see further sections of Chapter 6).

6.4. Analysis of the Semi-Empirical Model for Experimental Data

This section verifies whether the experimental data $C_d(\alpha)$ can be described by a simple semi-empirical model dependent on the geometric variable $x(\alpha) = \sin^2\left(\frac{\alpha}{2}\right)$. First, the form of the model is recalled, then the Weighted Least Squares (WLS) regression method is presented (with a full derivation in Appendix C), the quality of the fit is assessed, and the parameters (a, b) are visualised. Finally, physically meaningful parameter bounds are determined using the End-Points Pivot Rule, and the resulting boundary lines are presented in the graph C_d relative to $x(\alpha)$.

6.4.1. Model form

The model under consideration has a linear form with respect to the auxiliary variable;

$$C_d(\alpha) = a\,x(\alpha) + b, \qquad x(\alpha) \equiv \sin^2\left(\frac{\alpha}{2}\right), \tag{6.1}$$

where a is the slope parameter, while b is a free term interpreted as the base drag independent of α.

6.4.2. Regression method: weighted least squares (WLS)

Since the measurement uncertainties C_d are not homogeneous (heteroscedasticity), the weighted least squares method (WLS) was used. The weights were defined as the inverse of the variance [14];

$$w_i = \frac{1}{\sigma_i^2}, \tag{6.2}$$

where σ_i is the absolute uncertainty (here denoted by δC_d) corresponding to point i. The objective function and the definition of the weighted coefficient of determination R_w^2 are given in the Appendix C.

6.4.3. Goodness of fit-Experimental data

For the experimental data, the regression optimum WLS was obtained:

$$a^* = 0.6956, \qquad b^* = 0.4796, \qquad R_w^2 = 0.98879.$$

A high value of R_w^2 indicates that the linear relationship with respect to $x(\alpha)$ approximates the data trend very well in the studied angle range. A positive free term ($b^* > 0$) is particularly important for interpretation, as it indicates a non-zero drag even at small $x(\alpha)$, which is qualitatively consistent with the observation that drag does not disappear for slender geometries.

6.4.4. Parameter Space (a, b) heat map-Experimental data

In order to examine the sensitivity of the fit, a numerical scan of the (a, b) space was performed and the area $R_w^2(a, b)$ was calculated (Appendix C). The result is presented as a heat map in fig. 6.3. The contours (in particular, the level $R^2 = 0.95$) illustrate the area of parameters that give a very close fit, while the point (a^*, b^*) corresponds to the maximum R_w^2.

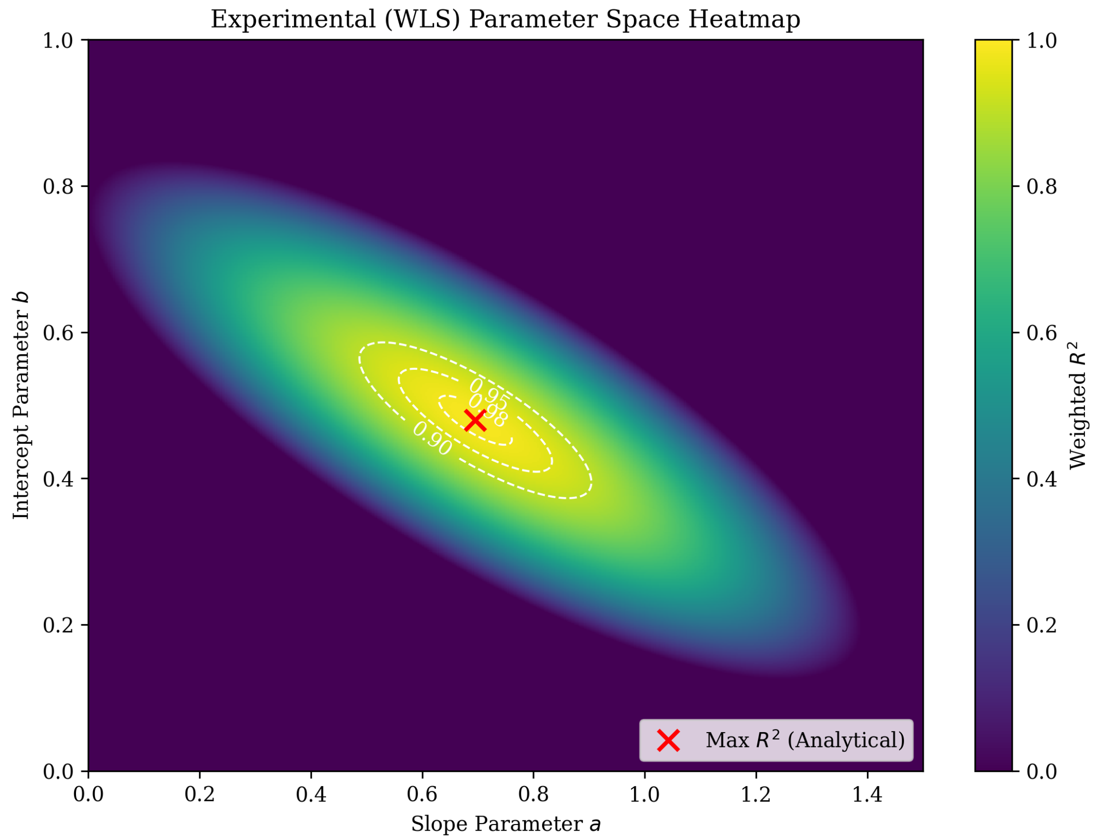


Figure 6.3: *Heatmap of the parameter space (a, b) for WLS regression (experimental data): the field $R_w^2(a, b)$ with contours, including $R^2 = 0.9$, $R^2 = 0.95$, $R^2 = 0.98$ and the optimum (a^*, b^*).*

6.4.5. End-Points Pivot Rule

Due to the small sample size ($N = 5$), traditional 95% confidence intervals based on the Student's t-distribution yield physically uninformative bounds; therefore, the End-Points Pivot Rule, described in detailed in Appendix C, was utilized to derive a rigorous parameter envelope based on local sensitivity. [14] It consists in 'rotating' the regression line around the weighted centroid $(\bar{x}_w, \bar{y}_w)$ and selecting the maximum and minimum slopes consistent with the most restrictive combinations of extreme points (left and right) along with their error bars. According to this rule, the following was obtained:

$$a_{\min} = 0.3450, \qquad a_{\max} = 1.0859.$$

The key principle of the pivot rule is as follows: after determining the new slope a_{new}, the free term should not be treated as an "anchor" point, but should be selected so that the line still passes through the centroid $(\bar{x}_w, \bar{y}_w)$:

$$\bar{y}_w = a_{\text{new}}\,\bar{x}_w + b_{\text{new}} \quad \Rightarrow \quad b_{\text{new}} = \bar{y}_w - a_{\text{new}}\,\bar{x}_w. \tag{6.3}$$

For the experimental data, the centroid (determined by weights $w_i = 1/\sigma_i^2$) is $\bar{x}_w = 0.4062$ and $\bar{y}_w = 0.7621$ (Appendix C), which gives the corresponding free terms:

$$b(a_{\min}) = 0.6220, \qquad b(a_{\max}) = 0.3210.$$

6.4.6. Fitting C_d to $x(\alpha)$ with pivot envelope

Figure fig. 6.4 shows experimental data C_d in the space $x(\alpha) = \sin^2\left(\frac{\alpha}{2}\right)$ together with error bars δx and δC_d and three straight lines: the optimal fit (a^*, b^*) and two boundary lines resulting from the pivot rule: $(a_{\min}, b(a_{\min}))$ and $(a_{\max}, b(a_{\max}))$.

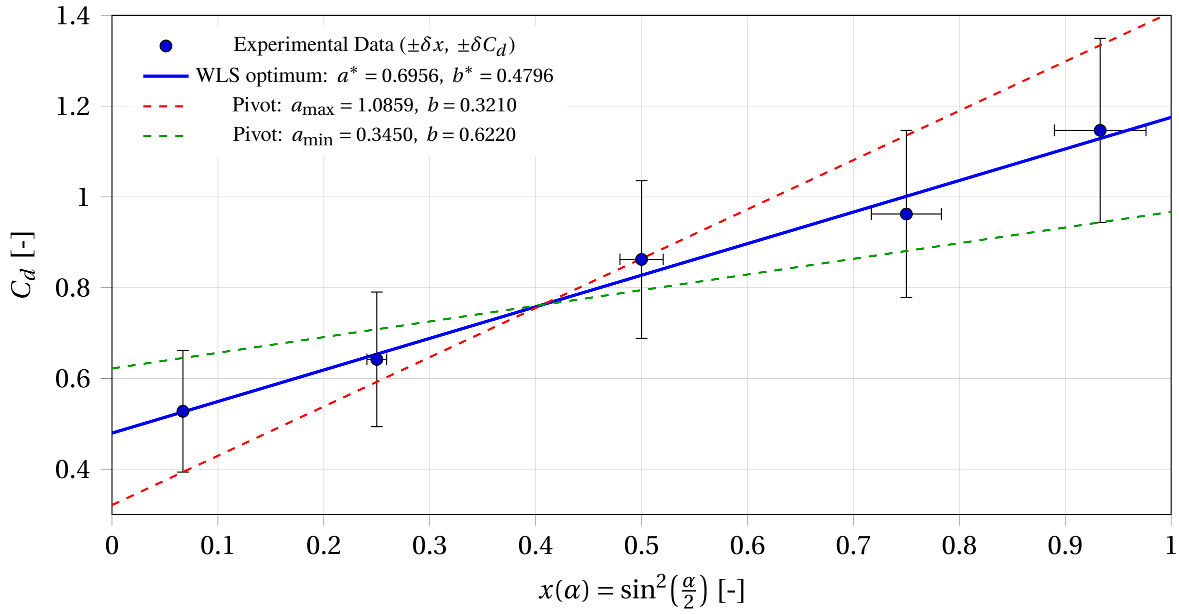


Figure 6.4: *Experimental data C_d relative to $x(\alpha) = \sin^2\left(\frac{\alpha}{2}\right)$ with error bars δx and δC_d, WLS fitting, and slope limits determined using the End-Points Pivot Rule method.*

6.4.7. Finalised form of the dependence $C_d(\alpha)$

Finally, the following WLS fit was adopted as the experimental model:

$$C_d(\alpha) = 0.6956 \sin^2\left(\frac{\alpha}{2}\right) + 0.4796, \tag{6.4}$$

where $a \in [0.3450, 1.0859]$ and $b \in [0.3210, 0.6220]$. The values $(a_{\min}, b(a_{\min}))$ and $(a_{\max}, b(a_{\max}))$ are treated as a physically meaningful envelope of parameter sensitivity resulting from the dominant measurement uncertainties.

6.5. Analysis of the Semi-Empirical Model for CFD data

In this section, only numerical data CFD obtained for the k-ω SST turbulence model (see Chapter 5) are analysed. The aim is to check whether the same form of the semi-empirical model, i.e. linear dependence on $x(\alpha) = \sin^2\left(\frac{\alpha}{2}\right)$, also describes the numerical predictions $C_d(\alpha)$. Unlike experimental data, CFD results are treated as deterministic: they are not assigned an uncertainty of the type δC_d, and therefore no parameter intervals (a, b) resulting from measurement errors are determined.

6.5.1. Regression method: Ordinary Least Squares (OLS)

For CFD data, the ordinary least squares method Ordinary Least Squares (OLS) (i.e., $w_i = 1$) was used, because for $C_{d,\text{CFD}}$ the statistical uncertainty of the points is not defined (no heteroscedastic σ_i). The objective function and analytical procedure are described in Appendix C. The use of OLS in this context is purely approximate: the fit is a tool for obtaining a concise description of the numerical trend, not for estimating parameter uncertainty. [14]

6.5.2. Goodness of fit-CFD data

For the CFD (SST) data, the following OLS regression optimum was obtained:

$$a^* = 0.8775, \qquad b^* = 0.3791, \qquad R^2 = 0.98139.$$

A value of R^2 close to one indicates that the linear relationship with respect to $x(\alpha)$ describes the numerical trend very well. At the same time, a positive free term $b^* > 0$ means that the CFD model also predicts a non-zero baseline drag level, i.e. it is not possible to achieve $C_d \to 0$ with decreasing $x(\alpha)$ in the range under consideration.

6.5.3. Parameter Space (a, b) heat map-CFD data

For completeness, the field $R^2(a, b)$ was also determined numerically in the parameter space (scan of the 500×500 grid) and presented as a heat map. It should be emphasised that in the case of CFD data, this map is not used to determine confidence intervals or parameter envelopes, as there is no information about point uncertainties (i.e. $\sigma_i \approx 0$ in the statistical sense). Consequently, the only substantively interpretable result is a single set of parameters (a^*, b^*) corresponding to the best OLS fit (see Appendix C).

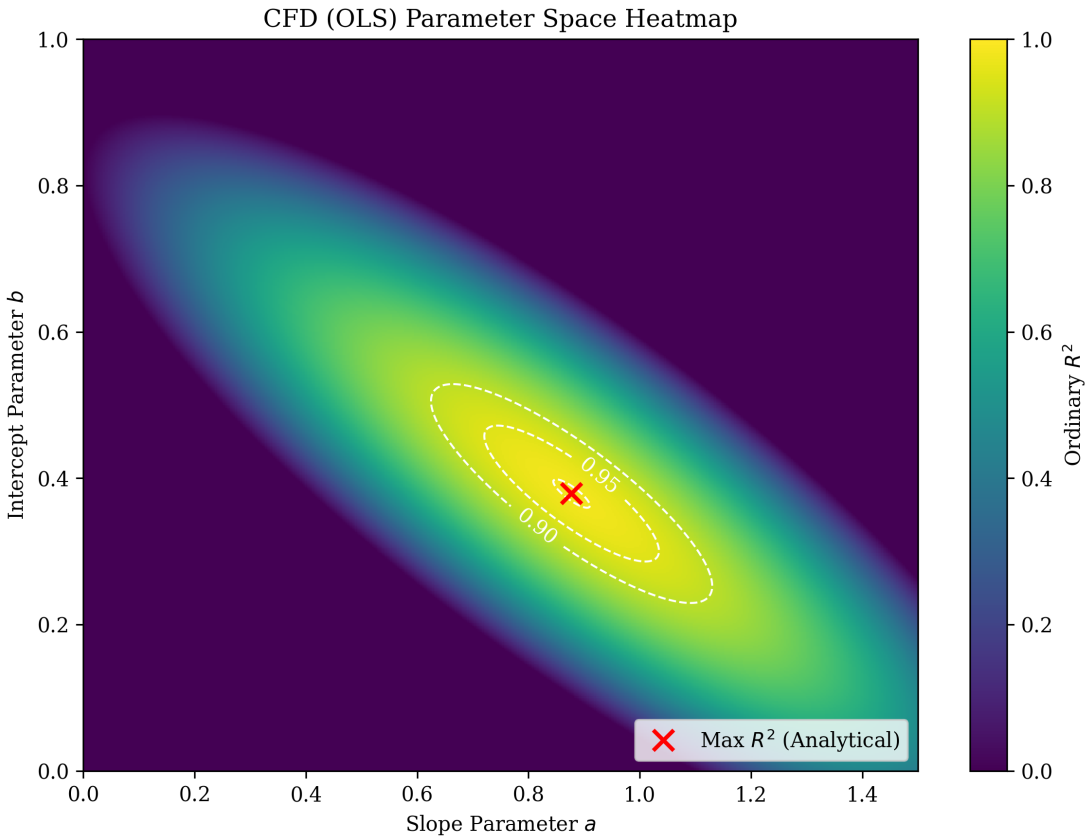


Figure 6.5: *Map of the parameter space (a, b) for OLS fitting to CFD data (SST): the field $R^2_w(a, b)$ with contours, including $R^2 = 0.9$, $R^2 = 0.95$, $R^2 = 0.98$ and optimum point (a^*, b^*).*

6.5.4. Fit of C_d to $x(\alpha)$ for CFD data

Figure fig. 6.6 shows the values of $C_{d,\text{CFD}}$ as a function of $x(\alpha)$ together with the trend line resulting from the OLS fit. Since the data are deterministic, no error bars of the type δx or δC_d are used.

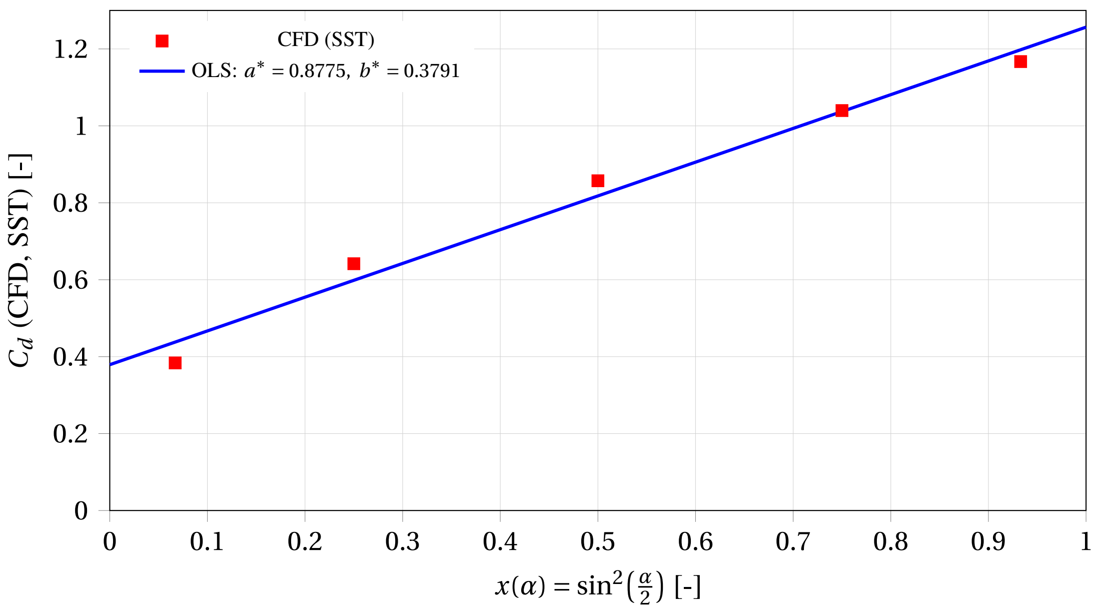


Figure 6.6: *CFD data (SST) C_d relative to $x(\alpha) = \sin^2\left(\frac{\alpha}{2}\right)$ and trend line from OLS fitting*

6.5.5. Finalised form of the dependence $C_d(\alpha)$ for CFD

Based on the OLS fit, the following concise description of the numerical trend was adopted:

$$C_{d,\mathrm{CFD}}(\alpha) = 0.8775 \sin^2\left(\frac{\alpha}{2}\right) + 0.3791, \tag{6.5}$$

which will be used in subsequent subsections for direct comparison of the parameters of the semi-empirical model for experimental and numerical data.

6.6. Comparison of Semi-Empirical Models for Experimental and CFD Data

This section summarises the parameters of the semi-empirical model fitted to experimental data (WLS, 6.4) and to CFD data for k–ω SST simulations (OLS, 6.5). The aim is to assess the consistency of the numerical trend with the physically measured trend, taking into account experimental uncertainty (described in section 3.4.3) and the sensitivity of the parameters estimated using the pivot method (Appendix C).

6.6.1. Summary of key parameters

For experimental data analysed using the WLS method, the following relationship was obtained:

$$C_{d,\mathrm{exp}}(\alpha) = 0.6956 \sin^2\left(\frac{\alpha}{2}\right) + 0.4796. \tag{6.6}$$

The fit coefficients are $a_{\mathrm{exp}} = 0.6956$ and $b_{\mathrm{exp}} = 0.4796$, respectively (see 6.4). According to the analysis using the End-Points Pivot Rule method, the slope a for the experimental data is physically consistent within the range:

$$a \in [a_{\min}, a_{\max}] = [0.3450,\ 1.0859], \tag{6.7}$$

which is a key consistency test for comparisons with the numerical model.

For CFD (SST) data, the OLS method yielded:

$$C_{d,\mathrm{CFD}}(\alpha) = 0.8775 \sin^2\left(\frac{\alpha}{2}\right) + 0.3791, \tag{6.8}$$

where $a_{\mathrm{CFD}} = 0.8775$ and $b_{\mathrm{CFD}} = 0.3791$ (see 6.5).

6.6.2. Comparison of parameters and validation argument (Pivot Check)

The slope value for CFD ($a_{\text{CFD}} \approx 0.88$) is greater than the experimental fit ($a_{\text{exp}} \approx 0.70$), which means that the numerical trend predicts a slightly higher bluntness sensitivity of the geometry, i.e. a faster increase in C_d with an increase in $x(\alpha) = \sin^2\left(\frac{\alpha}{2}\right)$. At the same time, the free term in CFD is lower ($b_{\text{CFD}} \approx 0.38$) than in the experimental fit ($b_{\text{exp}} \approx 0.48$), which indicates a lower level of base and skin drag in the numerical model for the limit $x \to 0$.

However, the key observation is that despite the difference between a_{CFD} and a_{exp}, the CFD slope satisfies the condition:

$$a_{\text{CFD}} = 0.8775 \in [0.3450,\ 1.0859].$$

This means that there is no statistical contradiction between the numerical trend and the experimental trend after taking into account measurement limitations. In other words, CFD describes the relationship $C_d(x)$ in a manner consistent with the 'corridor' allowed by the uncertainties of the experiment. Figure fig. 6.7 provides graphical confirmation of the consistency argument described above.

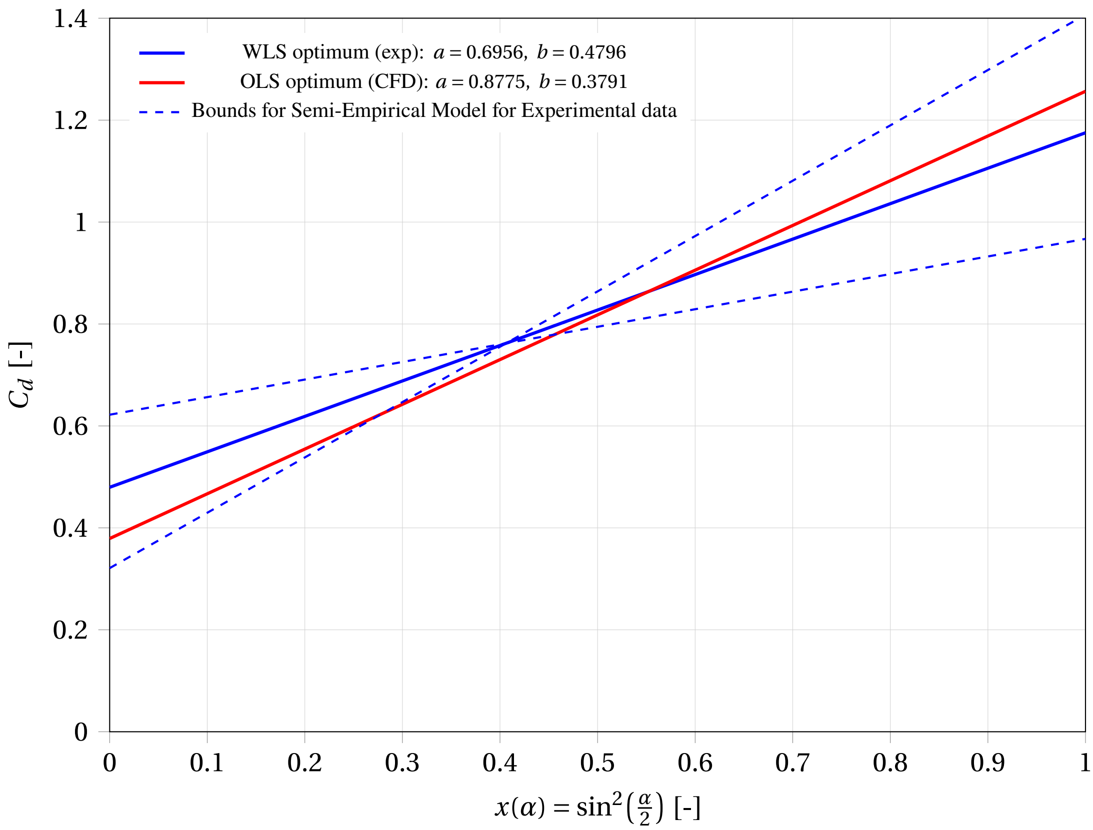


Figure 6.7: *Comparison of semi-empirical models in the space $x(\alpha) = \sin^2\left(\frac{\alpha}{2}\right)$: Experimental WLS fit and pivot envelope and CFD OLS fit.*

Despite differences in the best-fit parameters, the CFD slope falls within the experimental sensitivity range determined by the pivot method. This means that the numerical trend $C_d(x)$ is consistent with the measurements after taking into account the methodological limitations. Consequently, both approaches support the same physical interpretation: C_d increases non-linearly with the bluntness of the cone, and the positive free term indicates a non-zero component of the base drag.

6.6.3. Physical Interpretation of Parameters a and b

Within the adopted parametrisation, the coefficients provide distinct physical insights:

- a quantifies the *bluntness sensitivity*, defining the rate at which C_d increases as the cone becomes more bluff.
- b signifies the *base drag offset*, representing the residual drag component (viscous friction and persistent wake) that remains even in the limit of small x.

Consequently, the deviation of $(a_{\text{CFD}}, b_{\text{CFD}})$ from $(a_{\text{exp}}, b_{\text{exp}})$ reflects a shift in the balance between the base and bluntness-dependent contributions, arising from the contrast between the idealized numerical geometry and the actual, non-ideal experimental object. However, since a_{CFD} falls within the pivot range, these discrepancies do not undermine the consistency of the aerodynamic trends.

6.7. Verification of the dependence of the terminal velocity on the drag coefficient

In the considered range of Reynolds numbers of the order of $Re \sim 10^4$ (cf. Chapter 3), the flow around the cones is subsonic and dominated by separation and pressure drag typical for blunt bodies, and C_d can be treated as an effective quantity depending primarily on the geometry (including α) for a given flow regime. [6, pp. 75–89], [8] In this situation, the classic square drag regime applies, in which the resistance force is proportional to v^2, and the terminal velocity v_t results from the equilibrium $F_d = F_g$. [2]

The drag equation and the condition of equilibrium of forces yields:

$$v_t = \sqrt{\frac{2mg}{\rho A C_d}}, \tag{6.9}$$

which, with constant m, A and ρ, directly implies the relationship

$$v_t \propto \frac{1}{\sqrt{C_d}}. \tag{6.10}$$

To verify data coherence, an auxiliary variable was introduced:

$$x \equiv \frac{1}{\sqrt{C_d}}, \tag{6.11}$$

and the relationship $v_t(x)$ is presented in the form of a scatter plot. The relationship (6.10) predicts a linear progression $v_t = K x$ for the constant $K = \sqrt{2mg/(\rho A)} = 2.6952$.

The graph takes into account uncertainties in both axes:

- δv_t comes directly from the instrumental uncertainty of speed measurement (section 3.4.3),
- δx calculated from the propagation of uncertainty C_d in the transformation $x = C_d^{-1/2}$:

$$\frac{\delta x}{x} = \frac{1}{2}\frac{\delta C_d}{C_d}, \tag{6.12}$$

  which follows from logarithmic differentiation and is consistent with the propagation rules for power functions. [13]

The graph fig. 6.8 shows linear consistency across the entire data range. The strict linearity observed in the v_t versus $C_d^{-1/2}$ relationship empirically validates the assumption of the quadratic drag regime ($Re \sim 10^4$). The fact that the physical data rigidly adheres to this theoretical progression confirms that viscosity-driven scale effects are negligible for these kinematics, and that descent velocity is governed entirely by the geometrically determined pressure drag.

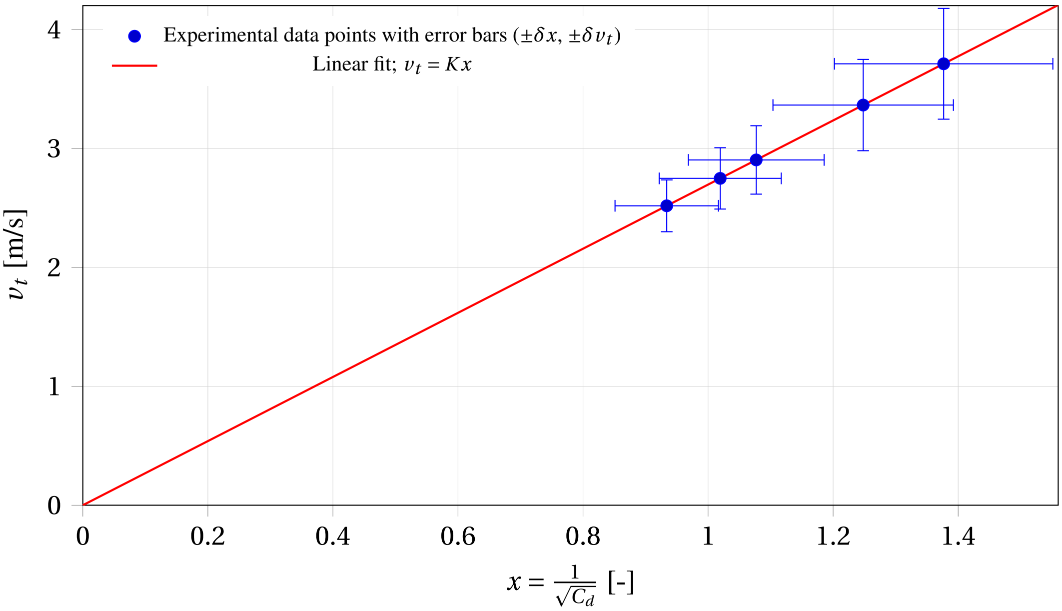


Figure 6.8: *Verification of the relationship* $v_t \propto \frac{1}{\sqrt{C_d}}$*: graph of* v_t *as a function of* $x = \frac{1}{\sqrt{C_d}}$ *with error bars* $\pm\delta x$ *and* $\pm\delta v_t$*.*

7
Conclusions and Recommendations

This final chapter synthesizes the primary findings of the research, addressing the objectives established at the beginning of this study. section 7.1 summarizes the experimental and numerical results, highlighting the physical mechanisms governing the aerodynamic drag of conical shells and evaluating the validity of the proposed semi-empirical models. Subsequently, section 7.2 outlines the inherent limitations of the current methodology and provides recommendations for future experimental and computational investigations.

7.1. Conclusions

The apex angle α is a key geometric parameter determining the aerodynamic drag of empty cones in free fall with their apex pointing downwards. Within the range of angles studied, a clear relationship was observed: as α increases (the cone becomes more 'blunt'), the terminal velocity v_t decreases, while C_d increases in a clearly non-linear manner, as detailed in the experimental findings of Chapter 4. For example, the 30° cone reached $v_t \approx 3.71\,\mathrm{m/s}$, while a 150° cone reached $v_t \approx 2.52\,\mathrm{m/s}$; at the same time, C_d increased from approximately 0.528 to 1.147.

The physical mechanism driving this resistance is dominated by form drag. The CFD analysis revealed that obtuse geometries generate a wide wake and extensive recirculation zones, which inhibit pressure recovery at the base. This results in a strong suction force that, combined with the stagnation pressure on the windward face, maximizes total drag. Furthermore, the results verify that within the experimental range, the terminal velocity follows the theoretical inverse-square root law ($v_t \propto \frac{1}{\sqrt{C_d}}$), confirming that the drag coefficient is independent of the Reynolds number for these sharp-edged geometries where flow separation is geometrically fixed (see Sections 6.3 and 6.7).

Mathematical Model. The simplified Newtonian model $C_d \propto \sin^2\left(\frac{\alpha}{2}\right)$ proved insufficient because it neglects the key physics of the wake, specifically the recirculation zone, as well as the underlying viscous contributions. According to the data analysis, the relationship $C_d(\alpha)$ is well described by a semi-empirical model:

$$C_d(\alpha) = a\sin^2\left(\frac{\alpha}{2}\right) + b,$$

where the experimental WLS regression yielded the specific relationship:

$$C_{d,\mathrm{exp}}(\alpha) = 0.6956\sin^2\left(\frac{\alpha}{2}\right) + 0.4796,$$

with a physically meaningful slope range resulting from measurement uncertainties: $a \in [0.3450,\ 1.0859]$.

The slope parameter bounds $a \in [0.3450,\ 1.0859]$, derived via the End-Points Pivot Rule, effectively encapsulate the experimental measurement uncertainty (see Section 6.4). The strictly positive intercept ($b > 0$) empirically confirms a baseline viscous and base-pressure drag component that persists even for highly slender geometries—a physical reality that Newtonian impact theory entirely overlooks.

For blunt cones ($60° - 150°$), the results CFD (SST $k-\omega$) were consistent with the measurements within the experimental uncertainty, exhibiting relative deviations of $\lesssim 2\%$ for three of the five tested geometries. This narrow margin rigorously validates both the experimental methodology and the selected computational configuration. At the same time, the semi-empirical trend for CFD:

$$C_{d,\mathrm{CFD}}(\alpha) = 0.8775\sin^2\left(\frac{\alpha}{2}\right) + 0.3791$$

is consistent with the ‘corridor’ allowed by the envelope of experimental data (see Section 6.6), which means that there is no contradiction between physical and numerical trends after taking into account measurement limitations.

A notable divergence was observed for the most slender geometry (30°). The experimental drag ($C_d \approx 0.53$) was significantly higher than the rigid-body CFD prediction ($C_d \approx 0.38$), representing an error of -27%.

This discrepancy is attributed to fluid-structure interaction. The 30° paper cone, being structurally less rigid than the obtuse models, experienced micro-vibrations and flutter during descent. These unsteady oscillations disrupt the wake and increase energy dissipation, leading to a higher effective drag than the idealized, rigid, and steady RANS simulation can predict. Sensitivity analysis using the more diffusive $k - \epsilon$ turbulence model yielded a higher drag coefficient ($C_d \approx 0.48$), suggesting that the real wake is more chaotic and extensive than the baseline SST model predicts for such slender shapes.

7.2. Recommendations for Future Work

The results apply only to hollow, thin-walled conical shells falling with their apex downwards in subsonic and approximately incompressible conditions, with only stable (quasi-stable) samples selected. Furthermore, the numerical analysis is based on stationary RANS simulations, which do not directly model the unsteadiness or flow-structure interaction relevant for very slender and flexible shells.

Building upon these limitations, the following recommendations are made for future work:

1. **Fabricate rigid test specimens:** utilize 3D-printed rigid shells rather than paper models to eliminate aeroelastic flutter. This would isolate the pure aerodynamic drag and likely reduce the discrepancy observed in the 30° case.

2. **Enhance temporal resolution:** Employ high-speed cinematography (increased FPS) to drastically reduce the dominant instrumental uncertainty (δv), thereby tightening the confidence intervals for the experimental drag coefficients.

3. **Conduct physical flow visualization:** Implement Particle Image Velocimetry (PIV) or wind-tunnel smoke tracing to physically validate the wake topologies and precise separation locations predicted by the RANS simulations.

4. **Implement unsteady numerical solvers:** Extend the computational methodology to Unsteady Reynolds-Averaged Navier–Stokes (URANS) to explicitly resolve the transient vortex shedding and aeroelastic wake oscillations inherent to highly slender geometries.

Additionally, a complete set of study-related files and data is available in the digital repository (Appendix D).

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Declaration of Generative AI Use

The author acknowledges the use of generative artificial intelligence tools during the preparation of this report to support specific technical and administrative tasks. **Microsoft Copilot** and **Google Gemini** were utilized for the following purposes:

- Assisting in the development and troubleshooting of Python code for data analysis;
- Refining the structural organization of the text;
- Generating LaTeX code for tables and formatting bibliographic entries (BibTeX);
- Creating code scripts for data visualization and charts;
- Compiling definitions for the glossary.
- Improving the wording of the report.
- Editing images and sketches.

The core analysis, engineering reasoning, and conclusions presented in this report remain the original work of the author. All AI-generated outputs were reviewed, verified, and edited to ensure accuracy and compliance with academic standards. The author assumes full responsibility for the content and integrity of this document.

A

Raw Experimental Data Log

This appendix presents the complete raw dataset for all 10 trials conducted for each cone geometry. The "Sequential Inclusion Protocol" described in Chapter 3 was applied: the first six trials exhibiting stable descent (verified by video analysis) were selected for the calculation of the mean terminal velocity (v_t).

Note on Data Exclusion: In instances where the specimen exhibited severe transverse trajectory divergence prior to passing the final fiducial marker (Point C), the transit time for the second interval (B-C) could not be optically resolved and is denoted as '-'.

Internal Apex Angle: 30°

Trial	Frames (A-B / B-C)	v_{mean} [m/s]	Status
1	**8 / 8**	**3.75**	**Included**
2	**8 / 8**	**3.75**	**Included**
3	**8 / 8**	**3.75**	**Included**
4	**9 / 8**	**3.53**	**Included**
5	8 / -	-	*Excluded (Trajectory Divergence)*
6	**8 / 8**	**3.75**	**Included**
7	**8 / 8**	**3.75**	**Included**
8	9 / 8	3.53	*Censored (Sample Quota Met)*
9	8 / 9	3.53	*Censored (Sample Quota Met)*
10	8 / 8	3.75	*Censored (Sample Quota Met)*

Internal Apex Angle: 60°

Trial	Frames (A-B / B-C)	v_{mean} [m/s]	Status
1	**9 / 9**	**3.33**	**Included**
2	**9 / 8**	**3.53**	**Included**
3	**9 / 9**	**3.33**	**Included**
4	**9 / 9**	**3.33**	**Included**
5	**9 / 9**	**3.33**	**Included**
6	**9 / 9**	**3.33**	**Included**
7	9 / 8	3.53	*Censored (Sample Quota Met)*
8	8 / -	-	*Censored (Sample Quota Met)*
9	9 / 9	3.33	*Censored (Sample Quota Met)*
10	9 / 8	3.53	*Censored (Sample Quota Met)*

Internal Apex Angle: 90°

Trial	Frames (A-B / B-C)	v_{mean} [m/s]	Status
1	**10 / 10**	**3.00**	**Included**
2	**10 / 10**	**3.00**	**Included**
3	**11 / 10**	**2.86**	**Included**
4	10 / -	-	*Excluded (Trajectory Divergence)*
5	**10 / 11**	**2.86**	**Included**
6	**11 / 11**	**2.73**	**Included**
7	**10 / 10**	**3.00**	**Included**
8	10 / 11	2.86	*Censored (Sample Quota Met)*
9	10 / 10	3.00	*Censored (Sample Quota Met)*
10	11 / 10	2.86	*Censored (Sample Quota Met)*

Internal Apex Angle: 120°

Trial	Frames (A-B / B-C)	v_{mean} [m/s]	Status
1	**11 / 11**	**2.73**	**Included**
2	**11 / 11**	**2.73**	**Included**
3	**10 / 11**	**2.86**	**Included**
4	**11 / 11**	**2.73**	**Included**
5	11 / 12	2.61	*Excluded (Kinematic Instability)*
6	**10 / 11**	**2.86**	**Included**
7	**12 / 11**	**2.61**	**Included**
8	11 / 10	2.86	*Censored (Sample Quota Met)*
9	11 / 11	2.73	*Censored (Sample Quota Met)*
10	12 / 11	2.61	*Censored (Sample Quota Met)*

Internal Apex Angle: 150°

Trial	Frames (A-B / B-C)	v_{mean} [m/s]	Status
1	12 / -	-	*Excluded (Trajectory Divergence)*
2	**13 / 12**	**2.40**	**Included**
3	**12 / 12**	**2.50**	**Included**
4	12 / 13	2.40	*Excluded (Kinematic Instability)*
5	**12 / 12**	**2.50**	**Included**
6	**12 / 12**	**2.50**	**Included**
7	**12 / 12**	**2.50**	**Included**
8	**11 / 11**	**2.73**	**Included**
9	11 / 12	2.61	*Censored (Sample Quota Met)*
10	12 / 13	2.40	*Censored (Sample Quota Met)*

B

CFD Convergence Plots and Mesh Details

This appendix contains the full set of visualization data supporting the numerical analysis in Chapter 5. It presents the velocity magnitude fields (wake topology), pressure distribution contours, computational grid, and convergence histories for each cone geometry.

For the 30° cone, results are presented for both the baseline k-ω SST model and the sensitivity study using the k-ϵ model.

30° Cone; Baseline $k-\omega$ SST Closure

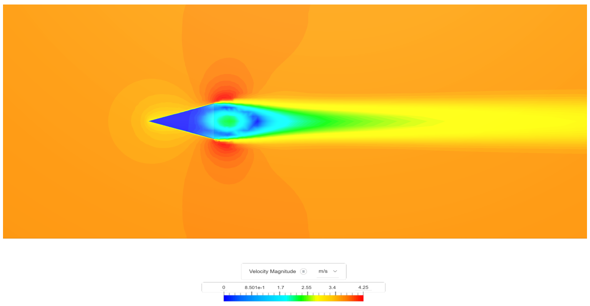

Figure B.1: *Velocity magnitude contour for the 30° cone (k-ω SST).*

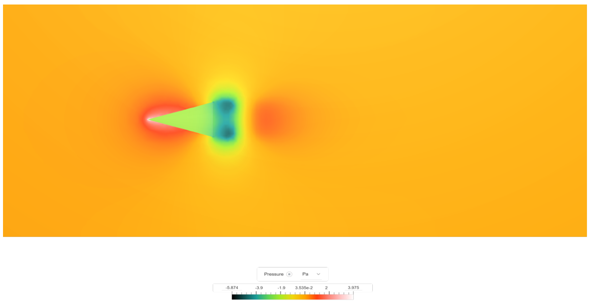

Figure B.2: *Pressure distribution coefficient (C_p) for the 30° cone (k-ω SST).*

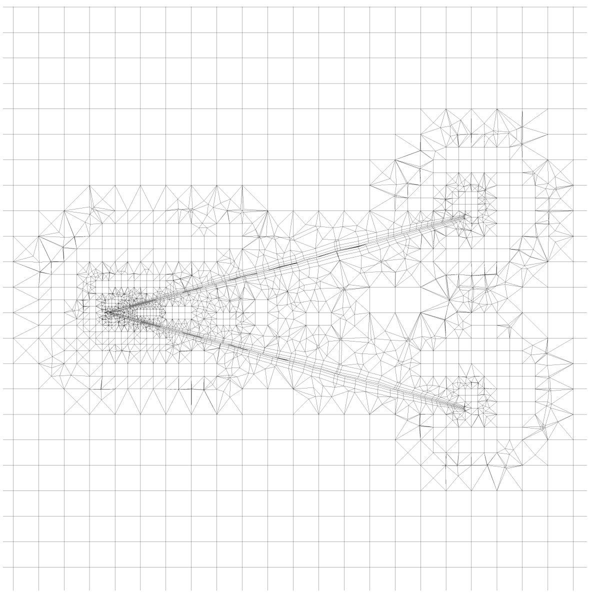

Figure B.3: *Computational grid details for the 30° geometry, showing the inflation layers near the wall boundary.*

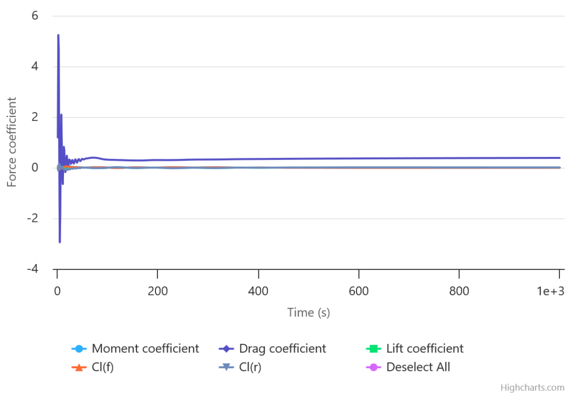

Figure B.4: *Convergence history of the drag coefficient (C_d) versus iteration number for 30° cone (k-ω SST). The asymptotic flattening of the curve demonstrates that a statistically stationary solution has been reached.*

30° Cone; Sensitivity Study $k-\epsilon$ Closure

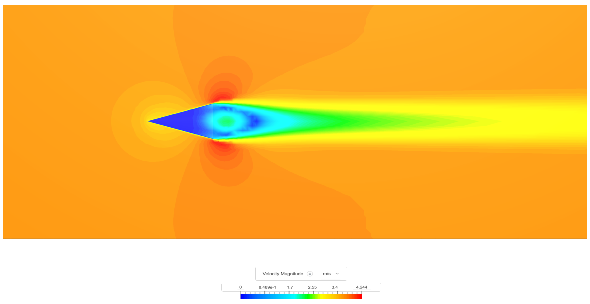

Figure B.5: *Velocity magnitude contour for the 30° cone using the k-ϵ model.*

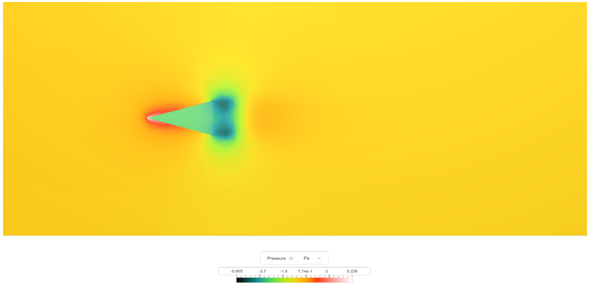

Figure B.6: *Pressure distribution for the 30° cone (k-ϵ model).*

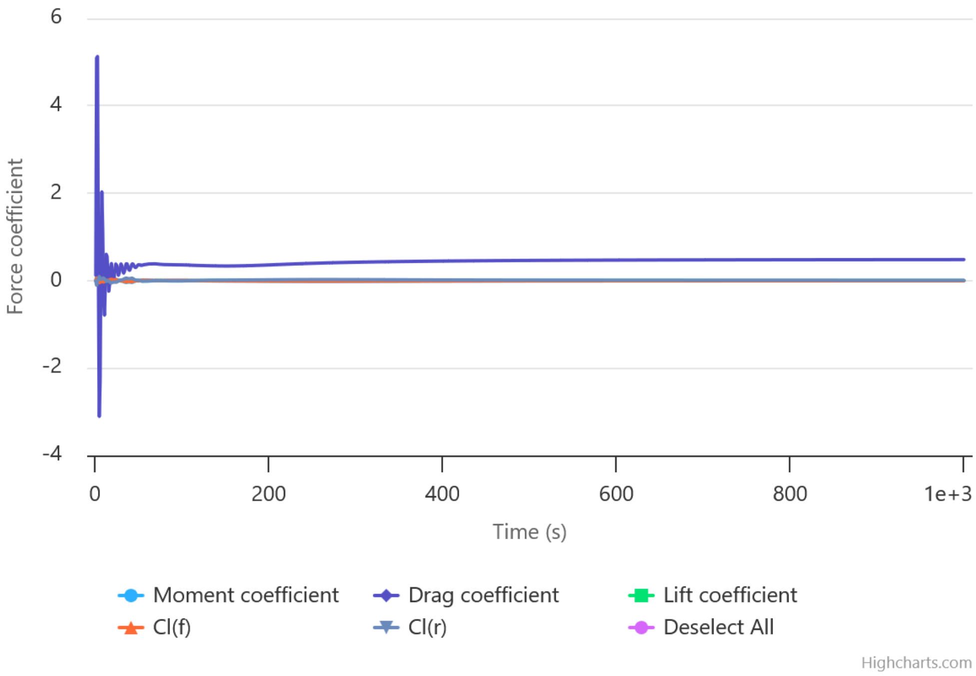

Figure B.7: *Convergence history of the drag coefficient (C_d) versus iteration number for 30° cone (k-ϵ model). The asymptotic flattening of the curve demonstrates that a statistically stationary solution has been reached.*

60° Cone

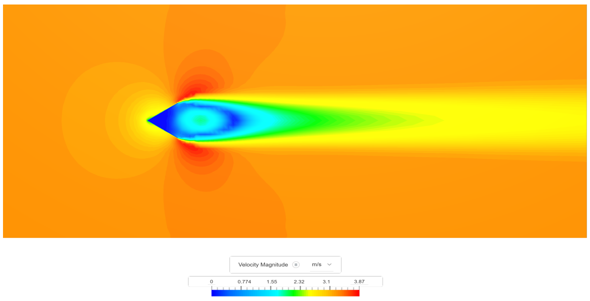

Figure B.8: *Velocity magnitude contour for the 60° cone (k-ω SST).*

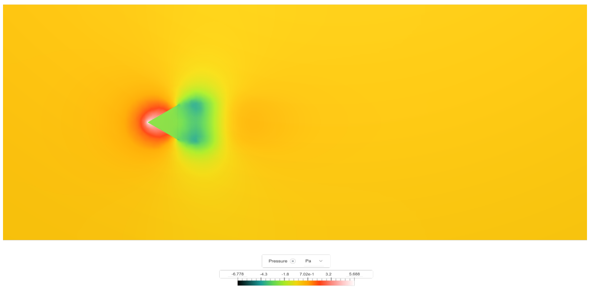

Figure B.9: *Pressure distribution on the windward and leeward surfaces of the 60° cone.*

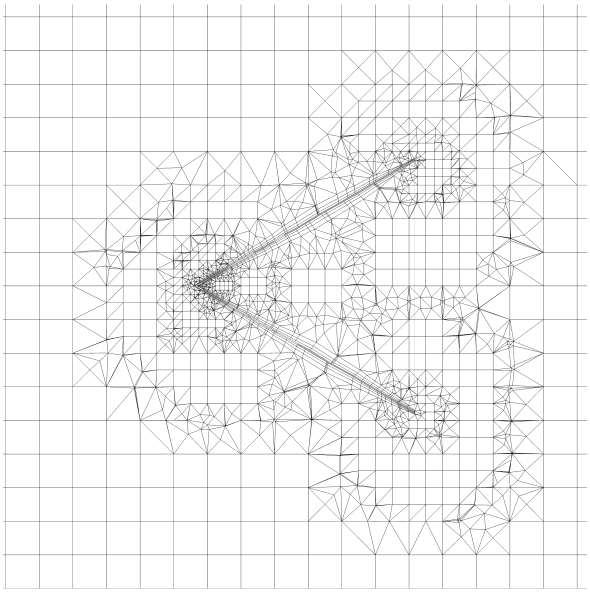

Figure B.10: *Computational mesh cut-plane for the 60° cone.*

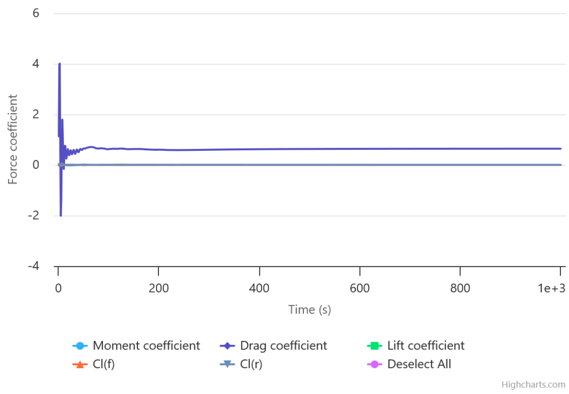

Figure B.11: *Convergence history of the drag coefficient (C_d) versus iteration number for 60° cone. The asymptotic flattening of the curve demonstrates that a statistically stationary solution has been reached.*

90° Cone

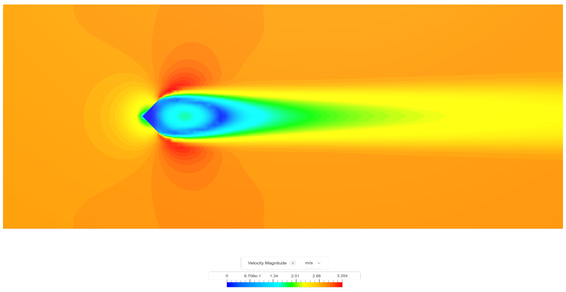

Figure B.12: *Velocity magnitude contour and wake topology for the* 90° *cone.*

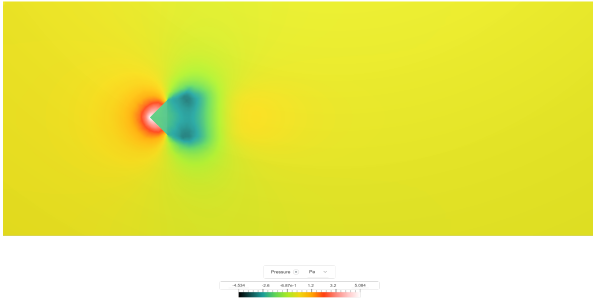

Figure B.13: *Pressure distribution for the* 90° *cone.*

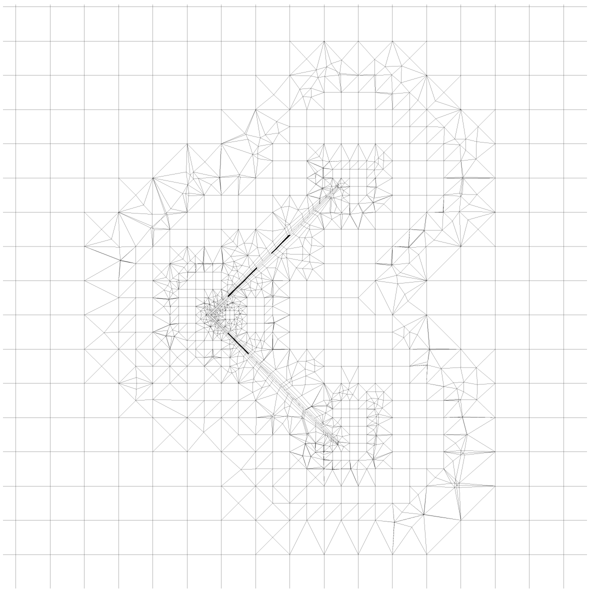

Figure B.14: *Computational mesh cut-plane for the* 90° *cone.*

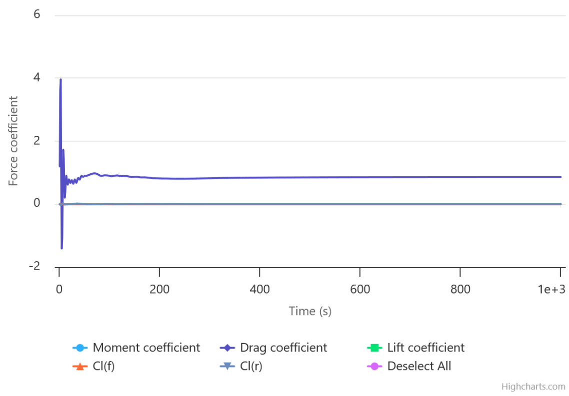

Figure B.15: *Convergence history of the drag coefficient (C_d) versus iteration number for* 90° *cone. The asymptotic flattening of the curve demonstrates that a statistically stationary solution has been reached.*

120° Cone

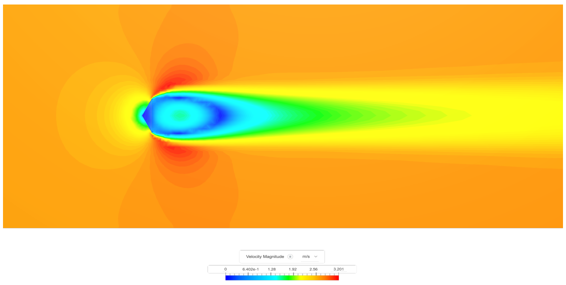

Figure B.16: *Velocity magnitude contour showing the widening wake of the* 120° *cone.*

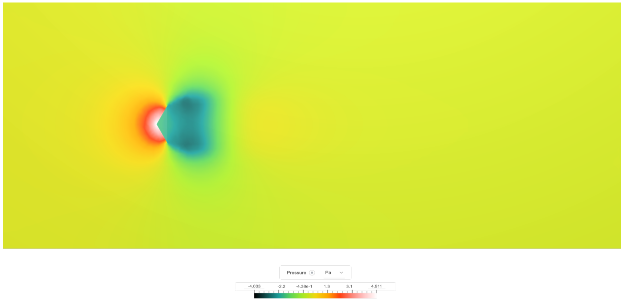

Figure B.17: *Pressure distribution for the* 120° *cone.*

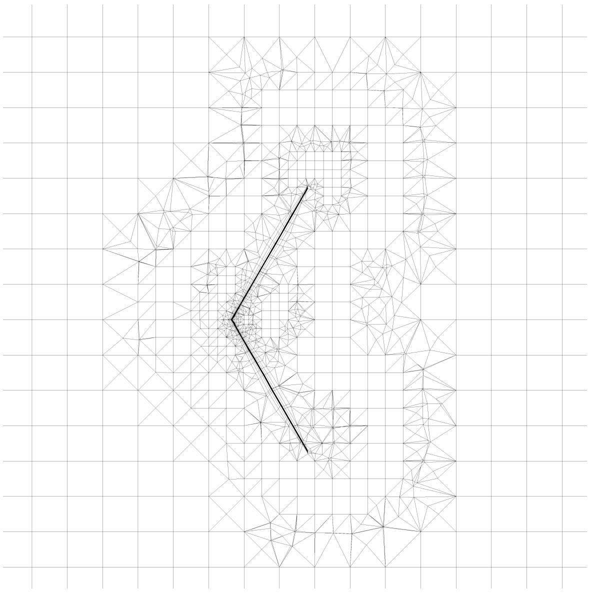

Figure B.18: *Computational mesh cut-plane for the* 120° *cone.*

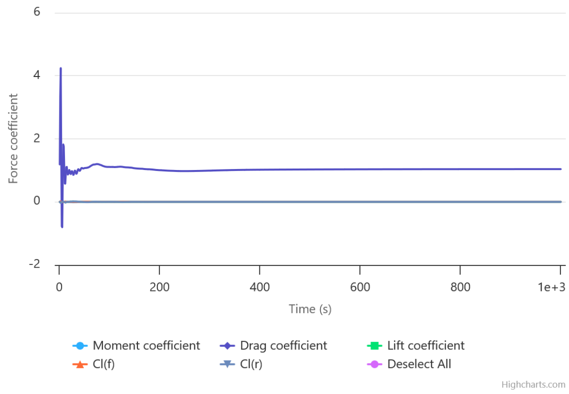

Figure B.19: *Convergence history of the drag coefficient (C_d) versus iteration number for* 120° *cone. The asymptotic flattening of the curve demonstrates that a statistically stationary solution has been reached.*

150° **Cone**

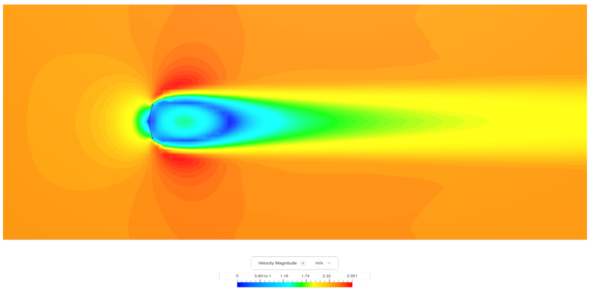

Figure B.20: *Velocity magnitude contour for the* 150° *cone, exhibiting the largest wake deficit.*

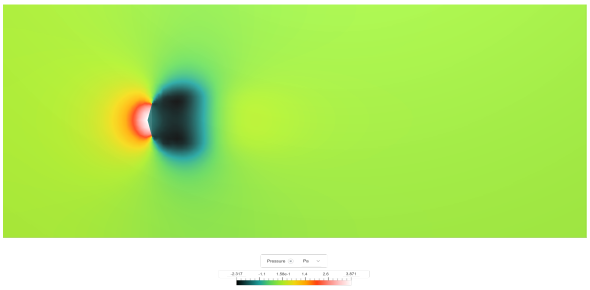

Figure B.21: *Pressure distribution for the* 150° *cone, showing maximum stagnation pressure.*

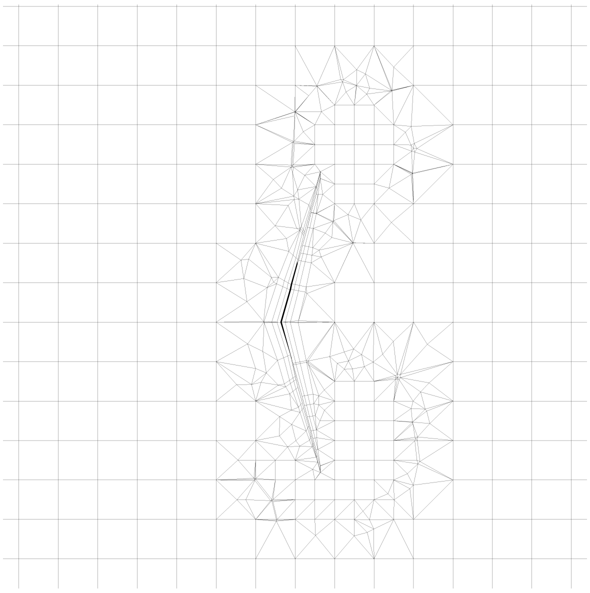

Figure B.22: *Computational mesh cut-plane for the* 150° *cone.*

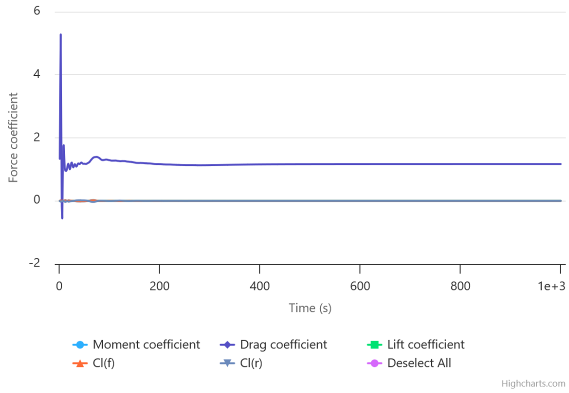

Figure B.23: *Convergence history of the drag coefficient (C_d) versus iteration number for* 150° *cone. The asymptotic flattening of the curve demonstrates that a statistically stationary solution has been reached.*

C
Regression Analysis and Parameter-Space Scanning

Methodological Overview

To determine the semi-empirical parameters a and b for the drag model $C_d(\alpha) = a\sin^2(\alpha/2) + b$, two distinct regression techniques were employed depending on the nature of the data [14]:

1. **Weighted Least Squares (WLS)** for experimental data, to account for heteroscedastic measurement uncertainty (σ_i).
2. **Ordinary Least Squares (OLS)** for CFD data, where no statistical uncertainty quantification was available.

Weighted Least Squares (Experimental Data)

Given the experimental dataset $\{(x_i, y_i, \sigma_i)\}$, where $x_i = \sin^2(\alpha_i/2)$ and $y_i = C_{d,i}$, weights were defined as $w_i = 1/\sigma_i^2$. The WLS objective function minimizes the weighted sum of squared residuals:

$$S_{wls}(a,b) = \sum_{i=1}^{N} w_i \left(y_i - (ax_i + b)\right)^2. \tag{C.1}$$

The goodness of fit was evaluated using the weighted coefficient of determination:

$$R_w^2 = 1 - \frac{\sum w_i (y_i - \hat{y}_i)^2}{\sum w_i (y_i - \bar{y}_w)^2}, \tag{C.2}$$

where $\hat{y}_i = ax_i + b$ are the model predictions and $\bar{y}_w$ is the weighted mean of the observed data.

Ordinary Least Squares (CFD Data)

For the CFD k-ω results, an unweighted OLS approach was used (equivalent to setting $w_i = 1$). The objective function simplifies to:

$$S_{ols}(a,b) = \sum_{i=1}^{N} \left(y_i - (ax_i + b)\right)^2. \tag{C.3}$$

Parameter Uncertainty: The End-Points Pivot Rule

To estimate the physical bounds of the slope parameter (a) for the experimental data, the "End-Points Pivot Rule" was applied. This heuristic determines the maximum and minimum plausible slopes by pivoting the regression line through the weighted centroid $(\bar{x}_w, \bar{y}_w)$ to the extremes of the error bars of the first (leftmost) and last (rightmost) data points.

The rigorous parameter bounds are established by calculating the slopes corresponding to the extreme bounding limits of the terminal data points:

$$a_{bound} = \frac{(y_{end} \pm \sigma_{end}) - \bar{y}_w}{x_{end} - \bar{x}_w}. \tag{C.4}$$

The minimum and maximum values from these calculated slopes define the plausible range $[a_{min}, a_{max}]$. This method provides a robust estimate of parameter sensitivity to the most constraining experimental uncertainties.

Numerical Implementation

The custom Python routine developed for this numerical analysis executes the following computational sequence:

1. Defines the parameter space grid (500×500 resolution).
2. Computes the R^2 scalar field for both WLS (Experimental) and OLS (CFD).
3. Extracts the optimal parameters (a, b) corresponding to the maximum R^2.
4. Calculates the experimental slope bounds using the End-Points Pivot Rule.
5. Exports the results to `.dat` files for visualization.

Note on Parameter Bounds: The slope bounds $[a_{min}, a_{max}]$ calculated by the script are physically meaningful only for the experimental data, where σ_i represents the measured uncertainty. For the CFD dataset, where no statistical uncertainty was quantified ($\sigma_i \approx 0$), the Pivot Rule collapses to the single OLS solution. Consequently, for the "bounds" output for the CFD case the code is set to return "skipped by configuration".

```python
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import os
os.chdir(os.path.dirname(os.path.abspath(__file__)))
#conf.

RESOLUTION = 500
A_RANGE = np.linspace(0, 1.5, RESOLUTION)
B_RANGE = np.linspace(0, 1.0, RESOLUTION)
A_GRID, B_GRID = np.meshgrid(A_RANGE, B_RANGE)

# Shared x values (sin^2(alpha/2))
x_vals = np.array([
    0.06698729810778929, 0.25, 0.5, 0.75, 0.9330127018922625
], dtype=float)

# helper functions

def compute_analytical_wls(x, y, weights):
    """Computes closed-form a, b for weighted least squares."""
    # Weighted means
    sum_w = np.sum(weights)
    x_w_mean = np.sum(weights * x) / sum_w
    y_w_mean = np.sum(weights * y) / sum_w

    # Numerator and Denominator for slope 'a'
    num = np.sum(weights * (x - x_w_mean) * (y - y_w_mean))
    den = np.sum(weights * (x - x_w_mean)**2)

    a_opt = num / den
    b_opt = y_w_mean - a_opt * x_w_mean
    return a_opt, b_opt

def compute_pivot_rule_bounds(x, y, sigma, weights):
    """
    Calculates a_min and a_max using the Pivot Rule (End Points Method).
    1. Finds weighted centroid.
    2. Uses only the leftmost and rightmost points to determine slope limits.
    """
    # 1. Calculate Weighted Centroid
    sum_w = np.sum(weights)
    x_cent = np.sum(weights * x) / sum_w
    y_cent = np.sum(weights * y) / sum_w

    # 2. Identify Extreme Points
    idx_min = np.argmin(x)
```

```
    idx_max = np.argmax(x)

    # Leftmost Point (L)
    x_L = x[idx_min]
    y_L = y[idx_min]
    sig_L = sigma[idx_min]

    # Rightmost Point (R)
    x_R = x[idx_max]
    y_R = y[idx_max]
    sig_R = sigma[idx_max]

    dx_L = x_L - x_cent
    dx_R = x_R - x_cent

    # 3. Calculate Limits based on End Points
    # Max Slope: Centroid to Top of Right OR Bottom of Left
    # Take the tighter constraint (minimum of the calculated max-slopes)
    # Slope to Top Right: (y_R + sig_R - y_c) / dx_R
    # Slope to Bottom Left: (y_L - sig_L - y_c) / dx_L  (dx_L is negative)

    slope_max_R = (y_R + sig_R - y_cent) / dx_R
    slope_max_L = (y_L - sig_L - y_cent) / dx_L
    # Since dx_L is negative, (y_low - y_c)/neg is a positive slope upper bound

    a_pivot_max = min(slope_max_R, slope_max_L)

    # Min Slope: Centroid to Bottom of Right OR Top of Left
    slope_min_R = (y_R - sig_R - y_cent) / dx_R
    slope_min_L = (y_L + sig_L - y_cent) / dx_L

    a_pivot_min = max(slope_min_R, slope_min_L)

    return a_pivot_min, a_pivot_max

def compute_r2_grid(x, y, weights):
    """Computes R2 surface over the global A_GRID, B_GRID."""
    y_w_mean = np.sum(weights * y) / np.sum(weights)
    ss_tot = np.sum(weights * (y - y_w_mean)**2)
    preds = A_GRID[:, :, np.newaxis] * x + B_GRID[:, :, np.newaxis]
    residuals = y - preds
    ss_res = np.sum(weights * (residuals**2), axis=2)
    return 1 - (ss_res / ss_tot)

def analyze_and_plot(name, x, y, weights, sigma, r2_grid, output_prefix, dat_filename,
    do_pivot_bounds=True):
    """Finds optima, extracts contours, plots heatmap, saves data."""
    print(f"\n--- ANALYSIS: {name} ---")

    # 1. Grid Optimum
    max_idx = np.unravel_index(np.argmax(r2_grid), r2_grid.shape)
    grid_a = A_RANGE[max_idx[1]]
    grid_b = B_RANGE[max_idx[0]]
    grid_r2 = r2_grid[max_idx]

    # 2. Analytical Optimum
    ana_a, ana_b = compute_analytical_wls(x, y, weights)

    print(f"Grid Max: a={grid_a:.4f}, b={grid_b:.4f}, R2={grid_r2:.5f}")
    print(f"Analytical: a={ana_a:.4f}, b={ana_b:.4f}")

    # 3. Pivot Rule Calculation
    if do_pivot_bounds:
     pivot_min, pivot_max = compute_pivot_rule_bounds(x, y, sigma, weights)
     print(f"Pivot Rule Bounds (End Points): a_min={pivot_min:.4f}, a_max={pivot_max:.4f}")
    else:
     print("Pivot Rule Bounds: skipped by configuration.")

    # 4. Plotting
    plt.rcParams['font.family'] = 'serif'
    fig, ax = plt.subplots(figsize=(8, 6))
```

```python

    # Heatmap
    im = ax.imshow(r2_grid, extent=[0, 1.5, 0, 1], origin='lower',
                   aspect='auto', cmap='viridis', vmin=0, vmax=1)

    cbar_label = r'Weighted $R^2$' if "Experimental" in name else r'Ordinary $R^2$'
    plt.colorbar(im, label=cbar_label)

    # Contours
    levels = [0.90, 0.95, 0.98, 0.99]
    contours = ax.contour(A_GRID, B_GRID, r2_grid, levels=levels,
                          colors='white', linestyles='dashed', linewidths=1)
    ax.clabel(contours, inline=True, fontsize=10, fmt='%1.2f')

    # Analytical Optimum (Red Cross)
    ax.plot(ana_a, ana_b, 'rx', markersize=10, markeredgewidth=2,
            label='Max $R^2$ (Analytical)')


    ax.set_xlabel('Slope Parameter $a$')
    ax.set_ylabel('Intercept Parameter $b$')
    ax.set_title(f'{name} Parameter Space Heatmap')
    ax.legend(loc='lower right')

    img_name = f"{output_prefix}_param_space.png"
    plt.tight_layout()
    plt.savefig(img_name, dpi=300)
    print(f"Saved plot: {img_name}")

    # 5. Save Data to .dat
    df = pd.DataFrame({
        'a': A_GRID.flatten(),
        'b': B_GRID.flatten(),
        'R2': r2_grid.flatten()
    })
    df.to_csv(dat_filename, index=False, float_format='%.6f')
    print(f"Saved data: {dat_filename}")


# 1.exp analysis (WLS)

y_exp = np.array([
    0.527576297466166, 0.6419620607599249, 0.862154655100411,
    0.9622421979824501, 1.146605134114744
], dtype=float)

sigma_exp = np.array([
    0.13379016000002, 0.14838547909538, 0.17356698767590,
    0.18439605617689, 0.20269902398342
], dtype=float)

# Weights = 1/sigma^2 for WLS
w_exp = 1.0 / (sigma_exp ** 2)
r2_exp = compute_r2_grid(x_vals, y_exp, w_exp)

analyze_and_plot(
    name="Experimental (WLS)",
    x=x_vals,
    y=y_exp,
    weights=w_exp,
    sigma=sigma_exp,
    r2_grid=r2_exp,
    output_prefix="exp_a_b",
    dat_filename="exp_a(0_1.5)_b(0_1)_param_space_heatmap_data.dat",
    do_pivot_bounds=True
)

# 2. cfd analysis (WLS with weight=1)
y_cfd = np.array([
    0.383698983, 0.64161545, 0.857249858,
    1.039745268, 1.16709476
```

```python
], dtype=float)

# OLS implies no uncertainty weighting
sigma_cfd = np.zeros_like(y_cfd)
w_cfd = np.ones_like(y_cfd)
r2_cfd = compute_r2_grid(x_vals, y_cfd, w_cfd)

analyze_and_plot(
    name="CFD (OLS)",
    x=x_vals,
    y=y_cfd,
    weights=w_cfd,
    sigma=sigma_cfd,
    r2_grid=r2_cfd,
    output_prefix="cfd_a_b",
    dat_filename="cfd_a(0_1.5)_b(0_1)_param_space_heatmap_data.dat",
    do_pivot_bounds=False
)
```

Listing C.1: *Python script used to compute WLS/OLS optima, Pivot Rule bounds, and generate parameter-space heatmaps.*

D

Digital Repository Access

In adherence to the principles of reproducible research and open science, the digital repository associated with this report provides unrestricted access to the complete underlying datasets, computational frameworks, and analytical source code. The repository is structured as follows:

- `/src` **– Source Code:** Python scripts utilized for data analysis, Weighted Least Squares (WLS) regression modeling, and automated plotting.
- `/data` **– Experimental Data:** Categorized datasets containing raw and processed free-fall kinematics, alongside uncertainty calculation matrices.
- `/cfd` **– Computational Visualizations & Data:** Post-processing visual assets, including mesh configurations, wake topologies, and pressure gradients, coupled with raw drag coefficient outputs for each tested conical geometry.
- `/results` **– Analytical Outputs:** Python-generated analytical data, encompassing synthesized parameter space files and final regression visualizations.

Scan the QR code below to access the repository:

Direct Link:
Aerodynamic Characterization of Hollow Conical Shells